



Heteroperception in joint action: Breakdown and reconciliation of collaborating learners' differing views of mathematical objects

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ABSTRACT

Reflecting across empirical results from three mathematics-education design-based research studies of collaborative problem-solving with material resources, we investigate social-interaction mechanisms by which action and discourse mobilize insight and conceptual change. We focus on moments when participants' multimodal utterances pertaining to features of the shared environment, or physical operations thereupon, gave pause to their peers, who apparently had been perceiving these features differently. Drawing on phenomenological philosophy of perception and cognitive-anthropological theories of joint action, we look to characterize why the materials' structural forms and quantitative-relational qualities became contested, how these emergent disagreements were negotiated, and what learning, if any, apparently resulted from reconciling perspectives. We present microgenetic analyses of three vignettes, one from each project, to offer our accounts of the social dynamics that engendered conflict and resolution. Our findings suggest that whereas divergent idiosyncratic perceptions may abound across students, these ambiguities surface and catalyze productive argumentation only when they translate into conflicting micro-operations, such as overt physical intervention that changes an object's location, orientation, or composition contrary to a peer's tacit intention. We comment on the socio-discursive affordances of different construction media and semiotic registers—embodied, material, inscriptional—as manifesting implicit heteroperception.

1. Introduction: situating an inquiry into heteroperception in a broader intellectual scope

We are exploring a proposed theoretical construct, heteroperception, that, we argue, lends insight into tacit social dynamics of collective problem solving, in particular as these bear on individual learning of mathematical concepts. *Heteroperception* depicts a socio-cognitive situation, in which a set of individuals—who are collaborating effectively in a joint endeavor involving sense-making and manipulating of common resources toward a shared goal—nevertheless are perceiving these resources differently. We are interested in multiple aspects of heteroperception—its philosophical plausibility, theoretical tenability, empirical expression, socio-material factors, interaction characteristics, and pedagogical repercussions—and to consider how these various aspects may cohere

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productively for the field of the learning sciences.

This introductory section begins by positioning our research interest—an educational inquiry into heteroperception—in the cognitive sciences' recent paradigmatic reconsideration of historical epistemological assumptions, that is, how embodiment may bear on heteroperception (Section 1.1). We then tie our educational inquiry into heteroperception to recent cognitive-science research on joint action, when several individuals are performing a task together (Section 1.2). Following, recognizing that when individuals engage in joint action they may perceive the situation differently, and that this heteroperception may lead to moments of confusion, we turn to seminal scholarship on the phenomenon of breakdown, to examine how it may illuminate the “life cycle” of heteroperception through the phases of implicit, explicit, and reconciled (Section 1.3). We then put forth our proposal for such a “life cycle,” which leads to our research questions that are designed to evaluate this proposal. At that point, the paper turns to its empirical Section 2, where the dynamics of heteroperception are laid out through three vignettes from our design-based research projects. Whereas these projects were conceived to build and evaluate pedagogical activities promoting educational practice, in turn they create empirical contexts for developing theoretical models concerning interesting phenomena, such as understanding heteroperception. A discussion in Section 3 will then draw out from the three projects unanticipated cross-cutting themes. In particular, our finding that group members may co-manipulate features of the learning environment despite their evident heteroperception dialogues in surprising ways with key tenets of sociocultural theory concerning the historical function of discourse in guiding the coordination of joint action. Citing cultural-historical activity theory (CHAT), we will submit that our study participants, who were contending with an “ahistorical” situation, namely a newly designed discovery-based pedagogical activity, implicitly understood what the overall *activity* is and what the basic *operations* were, even as they lacked initial consensus over the middle-ground *actions*. It is this lack of consensus, we will maintain, that precipitated the “life cycle” of heteroperception. The paper ends in Section 4 with our study's conclusions and implications.

1.1. Motivating educational inquiry into heteroperception

The very notion of *heteroperception* in joint action implies that our inquiry engages with the competing baseline assumption by which productive joint action requires identical perceptions of the situation across the set of participants. Such a baseline assumption, we submit, draws on traditional epistemological tenets that we reject—the idea that collaborative cultural practice, including semiotic interaction, is contingent on shared meanings (Russell & Whitehead, 1910). This perspective, grounded in Analytic Philosophy, we worry, denies or devalues a priori the prospects of effective collaboration under competing perceptions of a shared situation. The requirement of homogenous perception as a condition for interpersonal coordination would imply that heteroperception necessarily causes a Tower of Babel collapse of participants' coordinated reference to features of the environment, thus precluding effective joint action. An alternative view, from Continental Philosophy, shifts the unit of epistemological analysis from concept to practice (Wittgenstein, 1953). In parallel, cultural-historical perspectives on collective human behavior shift the unit of analysis from individuals to activity systems (Engeström, 1999; Leontiev, 1981). From these perspectives, the notion of heteroperception becomes naturalized.

The plausibility of heteroperception as a group phenomenon is further supported through studies of biological behavior. By this token, a team of students collectively confronting a confusing task is likened to a colony of foraging ants scoping a new terrain. We know that ants do well by embarking individually on random walks and, strikingly, by partially *miscommunicating* information on prized locations, to optimize for returns on collective exploration (Dreyer et al., 2025). More broadly, biological cybernetic studies would predict that aggregate solutions to novel problems emerge not through top-down dictation, delegation, or distribution of labor by some would-be leading authority but through bottom-up iterated interaction among multiple lower-level agentive components interacting with features of their environment (cf. Kauffman, 1995; Kelso, 1995; Thelen & Smith, 1994; van Geert, 1994).

We espouse an increasing shift to modeling human groups as complex dynamic systems in flux, from whence emerge and stabilize new cultural forms and norms (Dingemans et al., 2023). By this token, and extrapolating lessons learned from ants, a heuristic approach to human problem solving would be initially to: suspend judgment, language, and indexical precision; absorb in unpremeditated sensorimotor exploration; let embedded patterns reveal themselves through interaction; develop idiosyncratic perspectives; tentatively negotiate insights; and co-operate locally on mobilizing resources even in the absence of tight consensus over inherent meanings or global prospects of these local operations. Concepts can wait.

Foregrounding the role of sensorimotor exploration in conceptual learning is but one of several intellectual tributaries of the embodiment paradigm shift in the cognitive sciences (Newen et al., 2018). Indeed, this paper is situated intellectually in a Learning Sciences research program that is evaluating the purchase of embodiment on educational theory and practice (Abrahamson, Tancredi, Chen, Flood, & Dutton, 2023). Our rationale for this particular study is to take on an issue that has received considerable attention in educational research, collaborative learning, as an empirical context for furthering investigations of conceptual sociogenesis (Saxe, 2012). In similar vein, we are inspired by the idea of social epistemology as heteroglossia (Bakhtin, 1981) as well as models of collective engagement in a common task as soliciting heterogenous semiotic contributions that accumulate iteratively in a shared sub-strate informing subsequent action and discourse (Goodwin, 2002, 2013, 2018).

Notwithstanding the intellectual resonance of heteroperception with theories from a broad spectrum of the cognitive sciences, our disciplinary mandate as educational researchers compels us not only to document and characterize heteroperception as a phenomenon of collective action but also to query how heteroperception might ultimately bring about the accomplishment of curricular objectives, that is, how multimodal heteroglossia situated in group problem solving might converge toward learning disciplinary subject matter. As such, we first bolster the plausibility of heteroperception as well as its utility in investigating collective action by visiting research claiming that joint action need not imply shared perception: social groups' loosely-knit emergent functionality suggests that individuals engaged in co-operative action normatively may not be too concerned in monitoring how their peers are perceiving the

situation let alone what they are thinking (Subsection 1.2). One might then wonder how such functional dissensus could engender conceptual consensus. We therefore foreground the notion of breakdown as productively characterizing how and when unshared perception may mobilize social interactions leading to shared conceptual discourse (Subsection 1.3).

1.2. Joint action under heteroperception?

Appearances can be misleading. Imagine watching a group of people engaged in what appears to be a shared campaign. Imagine, moreover, that they are working in a shared spatial arena, even on one and the same artifact, for example assembling together a large structure. Observing these people, we will likely construe their activity as collaborative joint action, and we might assume that this action is undergirded by a shared perception of the situation. For, otherwise, we'd ask, how could their actions possibly be so well coordinated? Yet, questioning the necessity of attentional homogeneity in joint action could be important for a research agenda looking to theorize the emergence of conceptual consensus from collaborative problem solving. Could there be joint action without shared perception? Moreover, could there be joint attention without shared perception? Can we work together without realizing we are perceiving the situation differently?

Cognitive psychology research literature supports the theoretical notion of joint attention in the absence of homogeneous perception. The very idea of an ambiguous figure—Jastrow's proverbial duck/rabbit often comes to mind—implies interpersonal joint attention bearing heteroperception, where each person's attentional orientation to the shared phenomenon highlights different features that gravitate toward either one or the other Gestalt (Tsal & Kolbet, 1985). Still, when I see duck and you see rabbit, we are attending together.

Joint attention, a mark of human sociality, is a developmental achievement. The respective cognitive-developmental research of Tomasello (2019) and Melser (2004) reveals infants' landmark transition toward joint attention as they engage in joint action (see also Jacobson & Degotardi, 2022). These developmental theories are premised on the highly plausible assumption that adult-child interactions are asymmetrical: the adult perceives in a situation more contextually pertinent actionable structure (affordances) than does a child. Such an adult-child perceptual differential undergirds the foundational Vygotskian notion, the zone of proximal development (ZPD), where the adult responsively optimizes conditions for a child to engage a situation beyond their cognitive or motor means (e.g., Reed & Bril, 1996). Developmental echoes to Vygotsky's ZPD are found in cognitive-anthropological analyses of expert-novice (Goodwin, 1994) and teacher-student (Flood, Shvarts, & Abrahamson, 2020) interaction. Perceptual asymmetries have been documented also in the case of student-student interactions (Asterhan & Schwarz, 2009; Benally, Palatnik, Ryokai, & Abrahamson, 2022; Glachan & Light, 1982). As such, we might appreciate the possibility of joint action under conditions of heterogeneous perception. But what of would-be symmetrical groupings, such as a team of adults operating a system together?

A kneejerk theoretical stance would assume that joint action requires homogeneous perception. This assumption would reject as implausible the prospect of coordinated action in the absence of sufficient ontological and operational consensus about the things being acted upon. In a word, how could people act together on something, if they don't agree over what it is and what should be done with it? Such a view would implicitly rely on essentialist epistemology, by which objects constitute intact ontologies whose inherent meaning is revealed to individuals who can thus mentally represent this knowledge in equivalent form to align their understandings, plans, and actions. In contrast, alternative epistemological theorizations of coordinated meaning and action are rooted in scientific paradigms such as cybernetics that promote a different fundamental ontological sensibility, one that can embrace the looseness, emergence, and complexity of dynamical systems in flux (Kelso, 2021; Richardson, Washburn, Harrison, & Kallen, 2016). Such lines of cognitive-science research, we explain below, query traditional models of joint action, promoting instead a systemic theory whereby heterogeneous agents operating simultaneously on a shared set of resources can sustain functional dynamic stability in the absence of shared mental models.

Rejecting reductive theories of collective intentionality, philosopher Seemann (2024) develops an *enacted theory of joint attention*, where, “The openness of joint agents' intentions and meshing subplans is explained by appeal to their practical knowledge of how to maintain the process by way of which they pursue the collective intention” (p. 259; see also Crone, Gab, & Vincini, 2025; Rączaszek-Leonardi, Debska, & Sochanowicz, 2014). The would-be golden link between homogeneous perception and joint action is further interrogated by questioning the inter-cultural universality of previous assertions that were based on empirical work within WEIRD societies (i.e., Western, Educated, Industrialized, Rich, and Democratic; see Henrich, Heine, & Norenzayan, 2010). Investigating parent-child naturalistic interactions in non-industrialized societies, Taverna, Padilla, and Waxman (2024) demonstrated joint action in the absence of joint attention, thus posing questions for ontogenetic theories that foreground joint attention as a critical developmental marker (cf. Tomasello, 2019). In a resonant view from the philosophy of sports, Birse (2024) rejects representationalist assumptions that what is shared in joint attention is mental content. Instead, he sides with relationalist interpretations that look to include individuals' aspectual perceptual perspectives as well as the actions that perception inherently enables (see also Story, 2021, on the emergence of joint action without a theory of mind). Finally, Loehr (2022) problematizes what she views as facile models of individuals' sense of agency as mediating joint action.

We are at a point in this theoretical section where we have established the importance of educational research on collaborative joint action, and yet we have also asserted that participants in joint-action activities may be perceiving quite differently the objects they are co-manipulating. How, therefore, one might ask, could collaborative problem-solving foster shared knowledge? The closing subsection discusses breakdown as an epistemic pivot from implicit heteroperception to explicit disciplinary discourse.

1.3. Breakdown as epistemic pivot from heteroperception to conceptual consensus

Heteroperception may elude the researcher's eye until it causes communication breakdown. Here we first outline formative scholarship on breakdown as an *intrapersonal* phenomenology, and then we overview some further research on *interpersonal* breakdown, in particular as it pertains to conceptual learning.

The study of breakdown as an individual's personal experience prides an illustrious history (Heidegger, 1962). Koschmann, Kuutti, and Hickman (1998) survey the notion of breakdown through the respective lenses of three seminal scholars, as follows.

Heidegger develops the notion of breakdown by juxtaposition with ongoing non-reflective being. Tools and equipment—such as the keyboard by which this text is being input to an online word processor—are *ready-to-hand*, smoothly integrated into our mundane activities. However, when a breakdown occurs—a tool breaks, is missing, or proves inappropriate—its “invisibility” is disrupted. Perhaps the ‘e’ key is jammed on the keyboard. The resource then becomes *present-at-hand*, demanding our conscious attention and requiring us to reflect and interpret the situation.

Leont'ev postulates a hierarchical structure of activity consisting of—from top to bottom—activities, actions, and operations. Activities are composed of actions, actions are goal-directed, and operations are the unconscious adaptations that facilitate actions. Skilled performance involves chains of operations, each setting the stage for the next. Breakdown in this context occurs when conditions prevent the execution of an action. Previously routinized operations then surface to conscious actions that solicit intentional planning and decision-making. This unfolding of operations exposes the underlying structure of the activity and requires more conscious engagement with the process (see also Bernstein, 1996, Essay 5).

Dewey focuses on the organism's interaction with its environment. He rejects the Cartesian dualism of mind and body, asserting that organisms and their environments evolve together. A stimulus, in Dewey's view, is not an external force but rather the “total state of the organism” (LW 12:36, in Koschmann et al., 1998, p. 31) at a given moment. Breakdown occurs when acute disequilibrium in the organism's energy relations leads to its struggle for re-equilibration. This struggle is the basis for habit formation, as the organism consolidates past experiences to preempt recurring breakdowns. For humans, re-equilibration requires reflective inquiry, a controlled transformation of an indeterminate situation into a unified whole. Dewey points out that growth is achieved not by avoiding breakdown, but by engaging in inquiry to resolve the resulting disequilibrium.

The above survey of 20th-century philosophical inquiry into breakdown (Koschmann et al., 1998) is consonant with contemporary findings from the budding field of cognitive developmental psychology. Piaget (1971) theorized individuals' genetic epistemology as transpiring through cyclically accommodating mental schemes to assimilate features of the environment, allocating supplementary cognitive resources when these features present as obdurate (i.e., reflexive abstraction; see also Abrahamson, Shayan, Bakker, & Van der Schaaf, 2016). Drawing on enactivist philosophy, Pirie & Kieren (1989, 1992) likewise analyze mathematical learning as a process of *transcendent recursion*: a dynamic, recursive process, where individuals build upon prior skill to create increasingly sophisticated and abstracted understandings—constantly validating and reconstructing their knowledge by non-linear “folding back” movements through multiple levels of understanding, only to move forward again on more solid phenomenological grounds.

Whereas breakdown has been theorized primarily as occurring to individuals, thus *intrapersonal* negotiation, the construct of breakdown pertains also to dyads or groups, when apparent disagreement is surfaced and members negotiate across different perceptual perspectives on a given situation, hence *interpersonal* negotiation (Benally et al., 2022). As such, the very idea of breakdown at the social level assumes that the individuals had been engaging in joint activity without explicit consensus; the would-be consensus had been implicit “meaning as use,” where the meaning of a referent is its contextual function and application (Wittgenstein, 1953). We now turn to discuss interpersonal breakdown and its pedagogical horizons.

Cognitive anthropologists and conversation analysts view human communication as mobilized through rapid, frequent, iterated micro-breakdowns and resolutions (Goodwin, 2013). Breakdown is nothing dramatic in this sense. Rather, interlocutors constantly anticipate and repair breakdowns to maintain common referential ground (Isaacs & Clark, 1987; Schegloff, Jefferson, & Sacks, 1977). As such, communication scholarship normalizes rupture and repair as the constant, par-for-the-course, perhaps requisite state of affairs among interlocutors. Wherever people are engaged in discursive interaction, breakdown and re-attunement prevail, subtending a spectrum from slight misunderstanding through to coordination upsets. These insights into implicit mechanisms of human behavior “in the wild,” we will now explain, are valuable for leading organized interactions in classrooms.

Theoretical models of breakdown bear on the design of educational practice. Abrahamson and Wilensky (2007) exemplify activities that deliberately foster breakdown and steer its resolution toward conceptual learning by reconciling competing perceptions. Such productive negotiations can be facilitated through various media and activities, where a teacher may play an active role in optimizing the introduction of competing views (Abrahamson, 2009; Shvarts & Abrahamson, 2019). Breakdowns create opportunities for learning, when competing perceptions are “noticed, repaired, and reassembled in the form of new work practices or learning” (Ma & Hall, 2018, p. 511). By recognizing the developmental potential of disruptions and failures in intersubjectivity, educators can create more effective learning experiences that foster deeper understanding and promote the development of critical thinking skills (Mørch, Andersen, Eie, et al., 2024).

Benally et al. (2022) compare across several educational projects that each deliberately occasioned opportunities for dyads to collaborate across different yet conceptually complementary perspectives on a shared situation (e.g., investigating a body-scale polyhedron from within vs. without it). The researchers propose the construct of *conceptually generative perspectival complementarity* (CGPC) to capture the pedagogical potential of intentionally fostering interpersonal referential challenges among collaborating students. Their examples demonstrate how CGPC may lead either to: *mutuality*, when the two students maintain each their own perspective; *substitution*, when one of the students adopts the other's perspective; or *synergy*, when the two perspectives form a new and more elaborate understanding composed of, yet greater than the sum of the two perspectives (see also Wegerif, 2019).

1.4. Section summary, proposed model, and research questions

Acknowledging the prevalence of breakdown in social dynamics lends useful orientation for educational investigations of collaborative problem-solving by sensitizing researchers to evolutions, outbreaks, and potential dissolutions of heteroperception in their empirical data. It appears that education researchers should assume heteroperception as the default cognitive constitution among any set of humans engaged in a common activity, such as collaborative problem-solving. As individual students work together on a classroom open-ended inquiry activity, the surveyed literature predicts, they are expected each to perceive different embedded structures, enact different solution moves, and reason differently through transforming their immediate material environment. Analyzing such empirical data, researchers might infer from students' individual behaviors how they each perceive the situation. When breakdown occurs, these heteroperceptions then potentially lead to collaborative sense-making, articulation, negotiation, planning, and execution of technical operations in the shared domain.

Approaching joint action from enactivist and dynamic-systems theory perspectives appears important to educational research, because it orients data analysts toward accepting, even anticipating heteroperception in collaborative action. Still, *heteroperception is understudied, its processual mechanisms are under-theorized, and therefore its educational applications are under-utilized*. The objective of this paper is to investigate pedagogical dynamics of joint action under heteroperception. Our microgenetic analyses, reported herein, increasingly sensitized us toward multiple characteristics of pedagogically productive breakdowns, including social, material, and semiotic features of the learning environments. Building on the literature discussed above, and drawing on the cross-project emergent themes, we will be evaluating a proposed tripartite model of heteroperception process in collaborative problem-solving: (1) *implicit* (heteroperception is harbored unawares); (2) *explicit* (heteroperception is surfaced to individual and/or group attention); and (3) *resolved* (heteroperception is repaired, optimally resulting in new understandings that endorse both perspectives synergistically). The paper's rhetorical thrust is to present three vignettes, each from a different design-based research project investigating experimental pedagogical interventions, as case studies of our proposed model of collaborative conceptual bootstrapping through the evolution, breakdown, and reconciliation of heteroperception.

While demonstrating the productive qualities of heteroperception could inform educational theory of collaborative learning, it could also inform educational practice, by first allaying any didactic concerns over heteroperception in group activities. Moreover, our analyses might point educators to useful levers for anticipating, identifying, embracing, and shepherding perspectival conflicts toward the greater good. As such, we examined our case studies by asking:

1. What are the pedagogical potentials of heteroperception? Do these potentials evolve through predictable stages?
2. What factors of pedagogical situations bear on the pedagogical potentials of heteroperception?

Two housekeeping notes should orient the reader to this paper, the first on approach, the second on structure. We begin with a methodological apologia. As design-based researchers, our projects were by no means geared to follow golden standards of randomized control trials, even as the projects do arguably bear findings that could later warrant such quantitative methods (Goldin, 2000). Rather, we are looking to develop humble theory tethered to the contexts of observation yet with potential for generalization (Cobb, Confrey, diSessa, Lehrer, & Schauble, 2003). Moreover, because our projects all sprang from design problems, not research questions (Bakker, 2018), the two research questions, above, about the qualities and evolution of heteroperception, were not articulated a priori but only emerged through our laboratory's conversations. The construct of heteroperception itself came forth only *through* our analyses. As such, our "answers" to these a posteriori research questions are more in the manner of a perforated quilt stitched from occasioned opportunities to witness consistent behavioral patterns across the three projects' diverse contexts. To belabor this point, even as we stated our interests in the social, material, and semiotic dynamics of collaborative problem solving, no attempt was made—or likely could have been made—to isolate these facets for focused scrutiny. Second, given that the three vignettes are each from a different intact study, we have elected to precede each vignette by explaining its own background and methods, yet only as sufficient to contextualize the case studies for the purposes of this article, referring readers to earlier publications associated with each study. As such, we now dive directly into the case studies.

2. Case studies: comparative analysis of heteroperception and its effects across three design-based educational research projects

The three design-based educational research projects contributing to this article vary in their contexts of deployment, mathematical content, participant profile, and media resources. However, as we hope to demonstrate in this section, the projects yielded ontologically comparable phenomena as viewed through the micro-ethnographic analytic lens on individual cognition in social interaction. This section stages one vignette from each of the three projects in question: 3.1 The Icosahedron; 3.2 GRiD; and 3.3 PD by design. In turn, each vignette will highlight episodes evidencing the three successive phenomena of interest: (1) implicit heteroperception in joint action; (2) explicit heteroperception (i.e., the moment of awareness); and (3) consequences of heteroperception being acknowledged (i.e., cognitive, conceptual, or pedagogical implications of surfacing and discussing alternative perspectives on situated problem solving) (Table 1).

Table 1

Overview of the comparative study's data sources.

Project name	Context	Concept	Activity	Participants
Icosahedron	International educational research workshop at a university	3D geometry: polyhedra with a focus on Platonic solids	Assembling a body-scale icosahedron from minimal instructions	2 movement scientists, 1 sociologist, 1 learning scientist.
GRiD: Geometry Resources in Dance	Classroom-based design evaluation study in Bali, Indonesia	Basic shapes: circle	Forming a circle of four bodies with the use of a gridded mat	Four 11-year-old students. Indonesia
PD by Design: Teachers learn through creating and testing new resources	Pilot implementation of experimental PD intervention at a university	Basic arithmetic: division	Guiding student sense-making of division as an operation on objects	3 mathematics teachers, 1 peer serving in the role of a student. USA

As a final orientational comment for the reader, we stress that in parsing each vignette into the three proposed stages, certain judgment calls were necessary. When four people work together on an open-ended problem, their psychological experiences are not perfectly attuned nor their insights synchronized, making for “messy” analytic alignment. We thus chose, through our vignettes, to foreground the perceptual dramas of selected protagonists, as they advance through the stages, even as we report peripherally on the experiences of other dramatis personae.

2.1. Icosahedron

The *Icosahedron* activity invites four participants to construct from long wood dowels a body-scale voluminous geometrical object. The icosahedron (= twenty sides) is a polyhedron (= many sides) formed by twenty equilateral triangles (see Fig. 1A). It comprises 30 edges that meet at 12 vertices with 5 edges per vertex. At body scale, the form is so large that adults can step inside it (see Fig. 1B for the case of one adult, albeit our project's standing record is a total of 23 undergraduate pre-service teachers). Of note is that participants receive scarce instructions as to *how* to assemble the icosahedron, only a pile of dowels, pipe nodes for connecting the dowels at the vertices, and a small opaque printed image depicting the completed form (Fig. 1C). These limited process guidelines create conditions for participants to engage with an open-ended problem, while teachers and researchers can observe and investigate collaborative problem solving. For a detailed case study analyzing co-operative learning in this environment, readers are referred to [Palatnik and Abrahamson \(2026\)](#). The activity has been implemented in diverse settings for both educational and research objectives, including schools, pre/in-service teacher training sessions, graduate student seminars, and academic workshops, symposia, and conferences ([Rosenbaum, Palatnik, Reimer, & Abrahamson, 2024](#)).

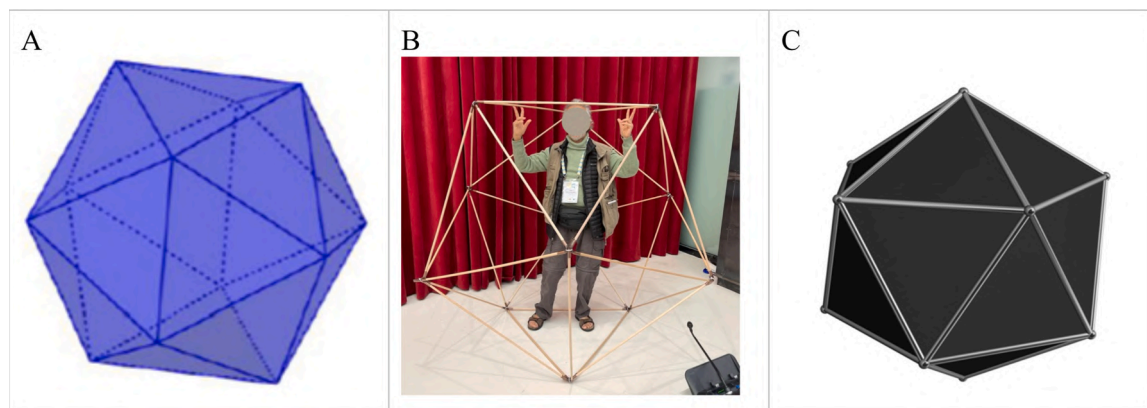


Fig. 1. The icosahedron: schematic image, body-scale model, and instruction image.

Note: Panel A: The icosahedron is a Platonic solid comprising 20 equilateral triangles. Panel B: The icosahedron at body-scale. Panel C: Students receive only an opaque image of the icosahedron, along with a pile of wood dowels for the edges and PVC connector for the vertices.

The vignette selected for this comparative study draws on data collected in the course of implementing the icosahedron assembly activity at an international workshop on learning analytics (EDaMMLA, 2024). Consequently, the four activity participants were researchers from disciplines across the cognitive sciences: two movement scientists, a learning scientist, and a sociologist. The data included audio–video footage both of the assembly process itself and of subsequent individual interviews with each participant (pseudonyms Barbara [B], Tanja [T], Filipe [F], & Helena [H]): semi-structured interviews (SSIs) with all four participants and additional stimulated-recall interviews (SRIs) with only Barbara (B) and Tanja (T), where they watched the video footage from the source activity and reflected on their experience during that activity; whom we selected due to their involvement in an episode of particular interest.

A methodological note is here due concerning the SRI technique. Weighing the scientific rigor of the clinical-interview as a methodological research instrument, Goldin (2000) urges us to separate the observed from the inferred and help readers distinguish among these two: “The goal must be to become as explicit as possible about the criteria for drawing particular sorts of inferences from particular sorts of observations” (p. 528). Indeed, critics of the technique's reliability have argued that when we ask a person to remember an earlier event, we must take their testimony with a large grain of salt, because they may be unwittingly generating false memories, implicitly led on by the interviewer's multimodal utterance (Loftus & Palmer, 1974). This “excavation fallacy” (Kordeš & Demšar, 2023) is an existential threat of qualitative analysis—the memory could be not only a false reconstruction but even an entirely new construction, because the interviewee had never been conscious of the experiences in question. We are facing a “can't live with it, can't live without it” crisis. Yet, as Guba and Lincoln (1989) put it:

The human is the instrument of choice for the constructivist, and, it should be stressed, the only possible choice during the early stages of an inquiry. Objections that humans are subjective, biased, or unreliable are irrelevant for there is no other option. (p. 175).

Hermeneutic and conversation-analysis approaches for purging dubious reconstructions to extract grains of truth can only go so far. Hopefully our SRI excerpts will evidence persuasive moments of ostensibly genuine surprise, when participants first become aware of an alternative Gestalt embedded in the environment. Finally, a semi-spineless support for SRI steps back to conceptualize the educational design-based researcher's role as developing learning scaffolds. If an interviewee arrives at conceptually advantageous insights through responding to an SRI protocol, then even if their recollections are fallacious, still the interviewer has fulfilled their instructional objective—clinical interviews are a powerful constructivist ploy (Ginsburg, 1997). Abrahamson, Trninic, Gutiérrez, Huth, and Lee (2011) explain how a design-based research team can keep improving their activity based on “what works” in clinical interviews:

[B]oth the interview protocol and the interactive affordances of the instructional materials evolved as we progressed through the pool of student volunteers. Most notably, we gradually incorporated into the protocol activities and prompts that the researchers initiated spontaneously during the interview and appeared to elicit student reasoning in line with the hypothetical cognitive construct we began developing. (p. 63)

In summary, while we acknowledge the tentative mettle of empirical data collected through clinical interviews, in particular stimulated-recall interviews, we appreciate their ultimate pedagogical utility. Beyond that, we shall leave it to the reader to glean the authenticity of our informants' testimonies.

The vignette presented in Fig. 2 tells a tale of four persons who, although successful in constructing a large complex geometric structure together, apparently each perceived it differently while doing so; they realized and reconciled their heteroperception not during the session itself but only subsequently, as individuals, when engaging in conversation with the researchers. For further details on the project's rationale, methods, and findings, readers are referred to Rosenbaum et al. (2024).

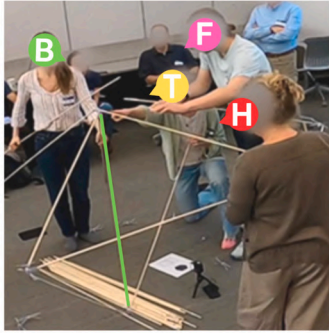
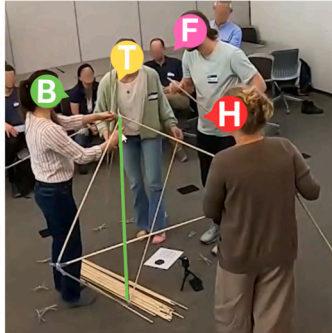
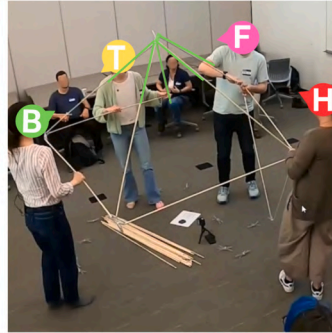
A Timestamp: 0:01:40  F: Ok, I'm making...[attempts to connect an edge to a vertex] T: But there is only..., is it... should be... B: But there's, yeah... T: I think we're...Is there five [points to a vertex]? I think there should be five vertices or five edges meeting. B: But then this one is extra, yeah [points to the green highlighted edge] T: Yeah.... B: We need to take this off [same edge].	B Timestamp: 0:01:52  B: [to herself] This extra one in the middle...[holds the green highlighted edge] . . . T: [looks carefully at what B is doing] . . .	C Timestamp: 0:02:30  B: Yeah, this is the, now, “top” [raising the top structure—see green highlighting]. T: Oh, cool. B: Yeah, we can think like this and then... T: Yeah. F: Yeah. B: ...we go from here. F: Alright. T: Ok. H: [silent, looks at “top”]
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Fig. 2. Transcript of selected key episodes in the source assembly activity.

Note: Participants' identities are anonymized with their pseudonym initials, with a “nose” indicating their gaze direction.

The one-minute vignette presented in Fig. 2 features selected episodes in the assembly activity video. We highlight these episodes, here, because they will serve, below, as conversation foci with the episode participants, during the stimulated-recall interviews that followed the assembly activity. First, T points out to the other group members that the vertex they are manipulating together already has five edges, so that F or B should not add any more edges (Fig. 2A). To this remark, B responds that the edge in the middle of the vertex (Fig. 2B) is superfluous, and subsequently she manipulates the vertex to remove one dowel, while the other group members watch carefully. Having completed that local structural alteration, B raises an intact structure (Fig. 2C) and calls it “top,” while the other group members apparently acknowledge this newly coined nomenclature either verbally and/or by gazing toward the term's referent.

During the individual SRI, T and B reflected on the assembly vignette that we have portrayed in Fig. 2. A set of figures, below, will feature T's and B's utterances from the post-activity individual SRI, supplemented by F's and H's utterances from the post-activity SSI. We offer these three post-activity interview episodes as demonstrating, per our thesis, three successive stages in the evolution of collective perception in joint action: (1) implicit; (2) explicit; and (3) resolved.

2.1.1. Stage 1: implicit heteroperception in joint action (icosahedron)

Following the assembly activity, T and B watched its video recording individually and commented on the moment presented in the vignette (Fig. 3). The transcript below Fig. 3, a compilation of their responses in the SRI, makes evident that T and B had been unaware of perceiving the structure differently, what we are calling implicit heteroperception.

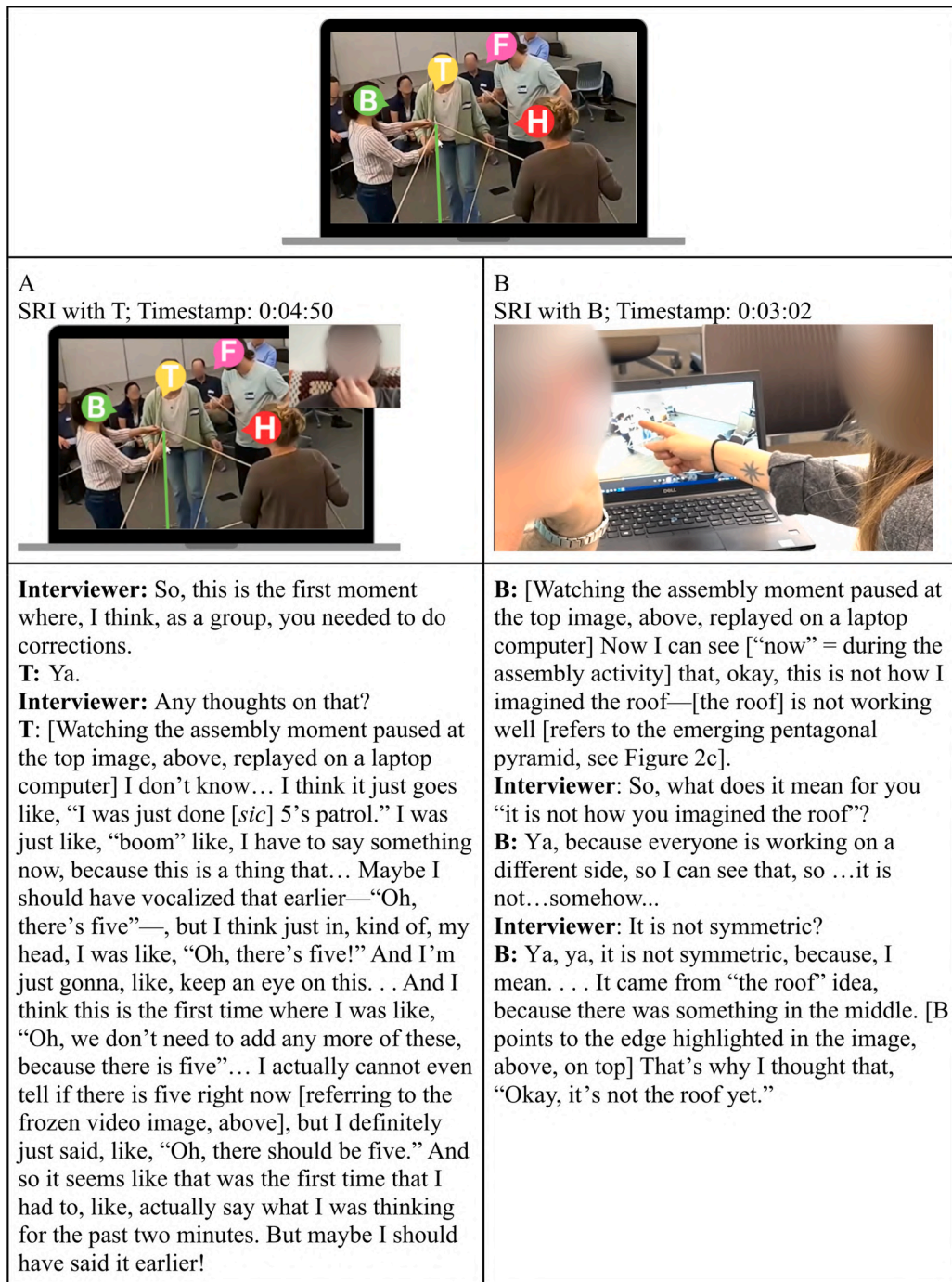


Fig. 3. Implicit heteroperception in joint action during the icosahedron activity.

Note: The figures show the SRI setting for T and B and T's and B's utterances as they recall their thoughts during the events transpiring in, and preceding the moment captured in the image at the top of this figure.

The above utterances during the individual post-activity reflections make evident participants' implicit heteroperception of the structure, both during the activity and then through to the interview. Curiously, the participants had engaged in effective collaborative construction of the structure, even as they were differently construing and imagining its emerging forms. Most prominently, whereas T was monitoring for "five" (five edges in each vertex), B tried to build "the roof" (a pentagonal pyramid). More strikingly, the stimulated recall interviews reveal that T and B bore these five-versus-roof heteroperceptions while discussing *and even contemporaneously manipulating* literally the very same edges.

We now turn to the post-assembly individual SSI with F and H, who said:

F: I think the roof idea or concept was established by B first. Somehow, I connect it to her. Then T had, I think, the idea of counting edges that arrive in the same vertex. And, I don't really remember what my idea was, to be honest, maybe because I abandoned it so early in favor of the other ones.

H: I remember, you know, that I don't have a clue how to do it for the majority of the time, but you know, F and others, they really behaved like someone who knows how to do it. So, I had a thing that I totally relaxed—"I trust them, it works." ... And this is also interesting, that this following of the group, it was like in harmony with my own thinking. It was not in conflict that "OK, I have to follow them and simultaneously, I have to find my own way," but it was, like, interweaving. So it was good.

The SSI further reveals that F was aware of T-and-B's heteroperception. Later in the interview, F disclosed that, for him, the activity's most difficult aspect was "knowing who[m] to listen to in such moments of confusion." Meanwhile, H did not always understand her peers' strategies for tackling the task at hand—she was comfortable contributing to the collective endeavor by following the group.

We thus witness heteroperception in joint action. And yet, we submit, was this action itself indeed joint as such? Not if we take "action" to connote a sentient agent's intentionality toward a particular imagined outcome, such as "five" or a "roof," in which case, the two participants were not truly engaged in joint action. At most, perhaps, all that was joint in their would-be joint action was a lower-order physical *operation*—attaching a particular edge to a particular node—an operation that constituted an overlapping component of their respective conflicting actions.

2.1.2. Stage 2: explicit heteroperception in joint action (icosahedron)

Fig. 4 will now make evident that it was only during the stimulated-recall interviews that T first became aware of the heteroperception she and B had implicitly borne earlier during the assembly activity.

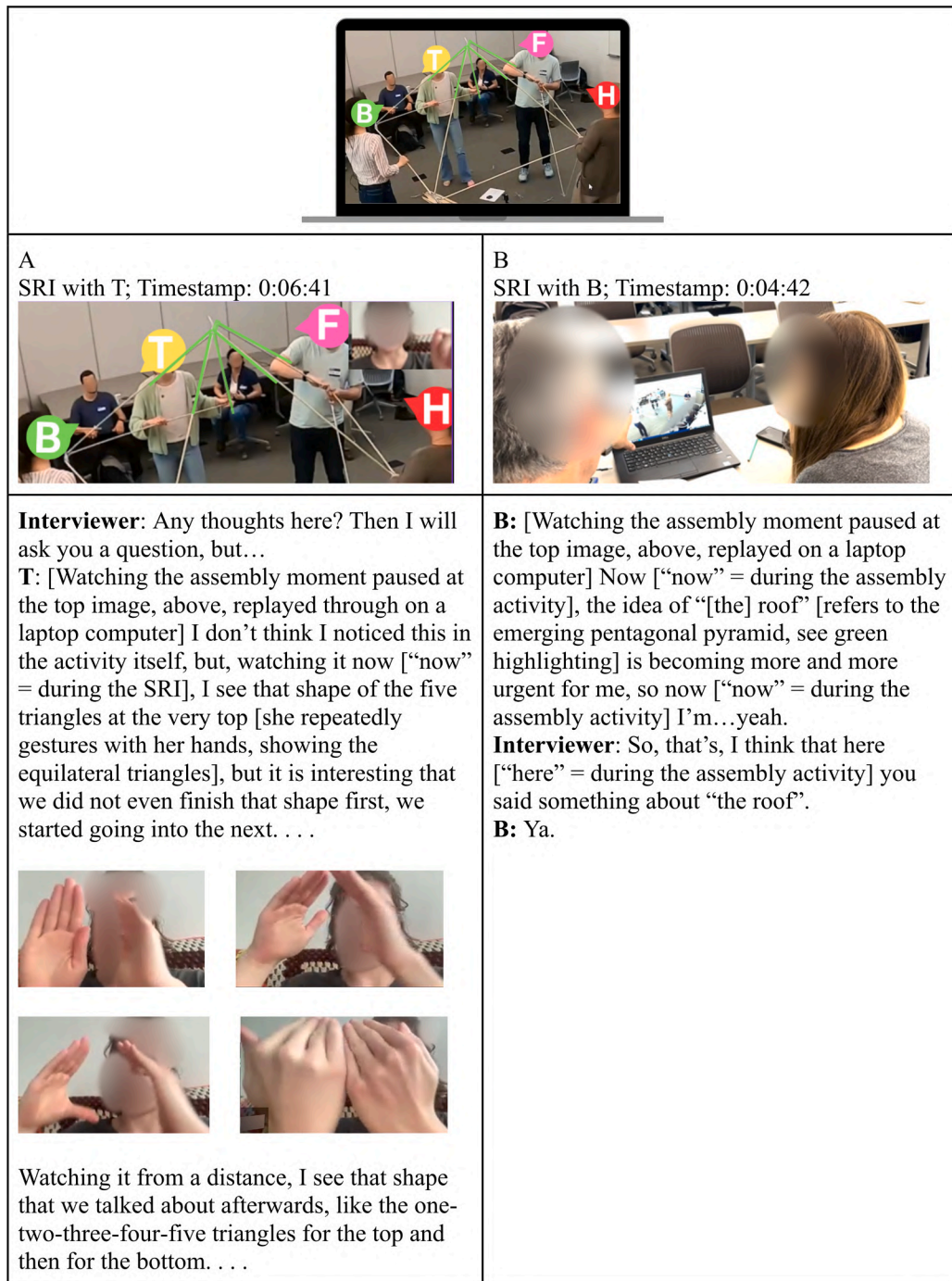


Fig. 4. Explicit heteroperception in joint action during the post-activity interviews.

Note: The figures show the SRI setting for T and B as well as T's and B's utterances, as they recall their thoughts during the events transpiring in, and preceding the moment captured in the image at the top of this figure.

For T, the phenomenon of heteroperception became explicit during the SRI, when she first perceived, in the video, the pentagonal pyramid that B was assembling. During her own SRI, B maintains the "roof" as the imagined gestalt she is materializing—she does not realize that at the same moment of construction, T was holding the same dowel and operating on the same vertex, but with a very different understanding and purpose.

In addition to the above comments from T and B during their SRI, F and H said the following during the SSI:

F: There were like different strategies presented, and I felt like there was this moment where we kind of were saturating, in terms of different ideas. So, I distinctly recall this moment where I just thought to myself, “Okay, now, I have to just step back and just choose whichever seems to be the dominant idea, besides what I think, and just follow that. Because, if I also come in with my idea and my plan, it is just gonna be a mess and we won't ever really agree on a decision.” And I think that was the most challenging part to me, to realize that. And then, once I realized, it suddenly became easier, ‘cause I just chose. I think it was T whose idea we followed in the end and also B's maybe, and then it was like, “Oh, yeah, now, now it starts to move on from, you know, A to B, yeah.” But there was this moment of confusion, where we didn't really know who[m] to listen to, kind of, and that was maybe the hardest part.

H: So, because my role was in reference to others, I didn't notice that something went wrong, okay? Because, you know, I didn't have this type of expertise or orientation. But I felt very happy that, “Ok, we have the problem, and they solved it.” It was very, like, I felt self-assured that, you know, I am following the right group, they know how to do it, and they solved it, and we are safe, and we continue. That was good.

F understood both T and B's different perceptual orientations and solution strategies, which led him to abandon his own strategy for the sake of task success and group harmony, doing what he could to promote group cohesion necessary for successfully completing the assigned task. Whereas H sensed that something was wrong with the construction (see Fig. 2), yet she did not realize that the other participants bore different perceptual orientations toward the emerging structure, nor did she bear her own idiosyncratic perception of the structure. H's perception of the solid was that of an artistic assemblage: “I think of this shape as the shape of a sculpture; it was more appealing to me, more than mathematics, you know, about the experience” (H, during the SSI).

2.1.3. Stage 3: resolved heteroperception in joint action (icosahedron)

Fig. 5 portrays T and B's utterances and reflections during the SRI concerning a moment in the activity video that affords a clear sight of “the roof”. The figure presents T's impassioned acknowledgment of her heteroperception and B's reflections through her prior perceptual knowledge.

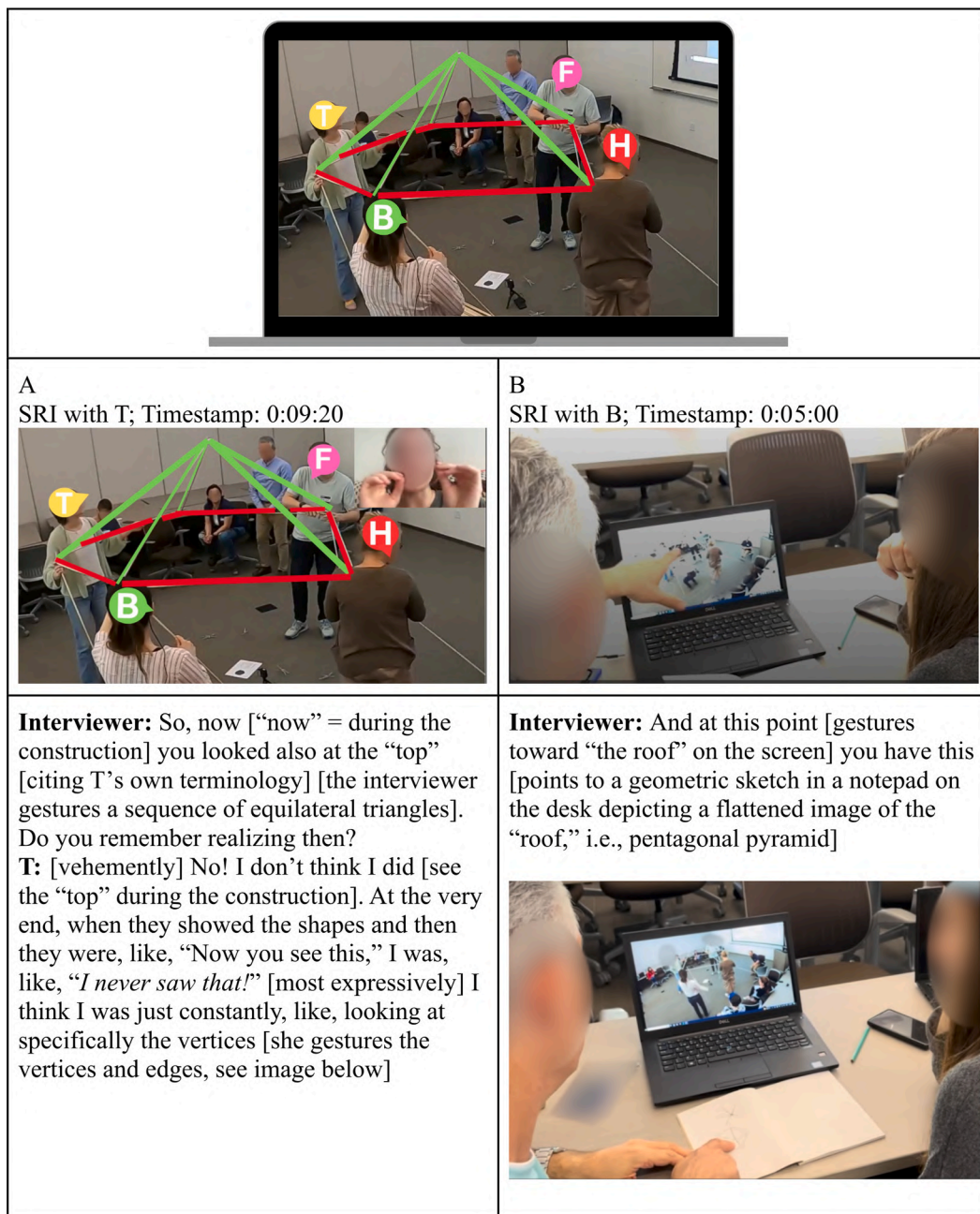


Fig. 5. Resolved heteroperception in joint action during the post-activity interviews.

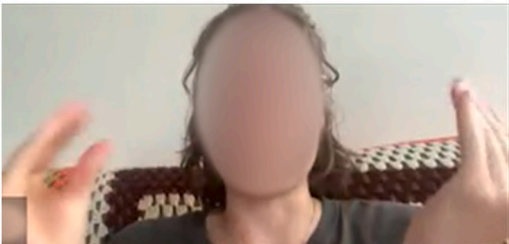
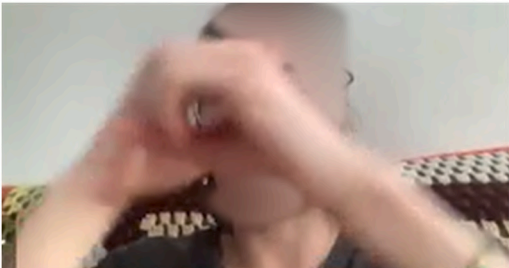
 I don't think I made out, like, the "bigger shape" until, like, the very end [she gestures "the bigger shape"]  'Cause my logic was, like, if we just keep an eye on these vertices, the shape will have to come together. I did not necessarily, like, move into like "ok, we gonna construct <i>this</i>  and then we gonna construct <i>this</i>	 B: Yes, yes, ya! Interviewer: Aha. Interviewer: And was it for you important to get also this structure [points to the flattened image of the "roof"]  or just this [points to a geometric sketch of "five" without the pentagonal frame of the "roof"]?  B: No. For the beginning, this ["five"]
--	--

Fig. 5. (continued).



Fig. 5. (continued).

Note T's emphatic exclamation, "I never saw that!" Appreciating the implicit heteroperception manifested during the team's joint action, T discerns and elaborates on additional properties of the icosahedron. She learns about the five triangles at the "top" and, most importantly, she first understands what B was trying to show the team during the construction. B, however, does not learn about the different perceptions of other group members. B maintains through to the SSI her construction strategy of "making the solid look like the one on the paper" (see Fig. 1C)—a strategy that still sustains her attention to "the roof."

In addition to the above comments from T and B during their SRI, F and H said the following during the SSI:

F: One of the two most enjoyable parts for me, or most rewarding parts, were the beginning, where there was nothing and it was very easy to show progress or to make some structures that would make us feel, "Ooh, yeah, actually, we kind of moved on quite a lot from the nothing state".

H: I would have never done it by myself from the very beginning, or it would have taken me lots of time! Yeah, it was very nice to do it in the team. And also this is like, you know, when you hear other people's comments, it really supports the metacognition, my own metacognition. Because they were faster, it was like hearing the voice just naming the process. That was, you know, I think it was a bit like the internal scaffolding.

F's remaining reflections concerned his pleasure at witnessing the team's initial socio-epistemic process of collectively devising potential structural formations and advancing with each participant's democratic and active contribution to the construction process. Whereas H had been comfortable imitating the peers' operations, per her report, comprehending the "5 s rule" prompted her sense-making of the actions that these operations were actuating. She stated, "You discovered that every point has like this five elements, and if I follow the rule that every corner has this five sticks, then I said, 'OK, if I have to repeat it, I know how to do it.'"

As a whole, the post-intervention interviews varied in their effect, across participants, in terms of the new realizations they ushered. Thinking about the pedagogical potential of collaborative tasks, we find that only through the SRI did T learn about the "roof," the 5 triangles "meeting" at the top, and, implicitly, about her heteroperception in relation to B. As T said in the SSI,

I think one thing that I noticed about myself is that the whole time I was very focused on, like, "that fiveness." Like the whole time I was just looking for fives. Like I was just counting fives and I never really imagined the whole shape ...

For B the situation is different. She did not realize, not even during the SRI, that T bore a different perception while the two of them were manipulating the very same vertex (see Fig. 2). Even though the five edges had served B, during the construction phase, as means of assessing the correct assembly of the imagined "roof" gestalt composed of five triangles, and even though B does point, during the SRI, to the five edges meeting at the top vertex (see Fig. 5B), still, the "five" constituted for B not an ends, as it had for T, but a means and to an ends and, therefore, as a necessary criterion. As B summarizes during the SSI:

Somehow I have this preconception of, for instance, uh, the surface. Each surface will be equal triangles, for instance. I already had this knowledge, so it was probably easy for me to visualize and to see what was working, and what was not working, when we created the shape.

Fig. 6 shows the three successive phenomena of interest, as they occurred in this case. Fig. 6A (implicit heteroperception) shows that, during the analyzed vignette, the four participants bore heteroperceptions of the construction (even while manipulating the same geometric element). T counted 5 edges, independent of the angle between them. B built a "roof" composed of 5 equal triangles. F understood that T and B held these heteroperceptions and decided not to further develop his own strategy. H followed the group without a strategy of her own nor appreciating the implicit heteroperception. Fig. 6B (explicit heteroperception) captures moments during the post-activity interviews, when the participants were invited to reflect on the collaborative construction. T first acknowledged her heteroperception relative to B by suddenly apprehending the structural referent of "roof" or "top" that B had been alluding to during the construction phase. In contrast, B had already been attending to the "five," which T had championed, as part of her own "roof" gestalt. B never realized, not even during the interviews, that she was in heteroperception with T at that moment in the collaborative construction. F and H each sustained their earlier understandings of the situation, while discussing it with the interviewer. Finally, Fig. 6C shows the partial resolution of the heteroperception as we understood it from the SRIs and SSIs, namely: T, B, and F had been in heteroperception while building "the roof." However, they now can all perceive "the roof"—composed of five equilateral triangles with five edges meeting at the top vertex—as a strategic goal component for building an icosahedron. H continued to trust and follow the group's strategy, unaware of the heteroperception. H briefly mentioned "the rule" that "every corner has this five sticks".

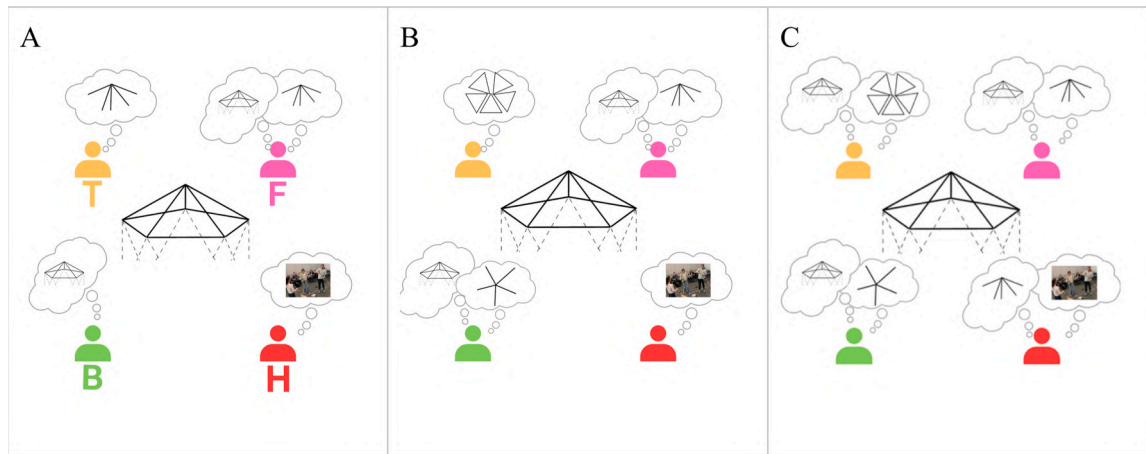


Fig. 6. Icosahedron construction activity: Three stages in the evolution of heteroperception.

Note: Panel A: Implicit heteroperception in joint action; Panel B: Explicit heteroperception; Panel C: Resolved heteroperception.

2.2. GRiD

Geometry Resources in Dance (GRiD, [Apsari & Abrahamson, 2026](#)) is a geometry learning activity emulating principles of the action-based genre of embodied design ([Abrahamson, 2014](#)). As such, the activity unfolds in two steps: (1) a set of assigned movement tasks occasion opportunities for students' to spontaneously generate or activate *attentional anchors* ([Hutto & Sánchez-García, 2015](#)), which are perceptual resources that support coordinated motor engagement with the environment; and then (2) symbolic artifacts interpolated into the working pace offer utilities for students to enhance their movement strategies by adopting the artifacts as frames of reference and action, yet in so doing the students shift to a mathematical discourse ([Abrahamson et al., 2023](#)). GRiD is a unique embodied design, in that it instantiates students' authentic cultural-historical practice, namely Balinese dancing, along with its tacit spatial-orientation matrices ([Apsari & Abrahamson, 2025](#)). As a geometry activity, GRiD's objective is for students to appropriate a particular diagrammatic scheme, displayed on a mat (see [Fig. 7A](#)), as bearing utilities for managing the personal and interpersonal coordination of traditional dance postures and movements.

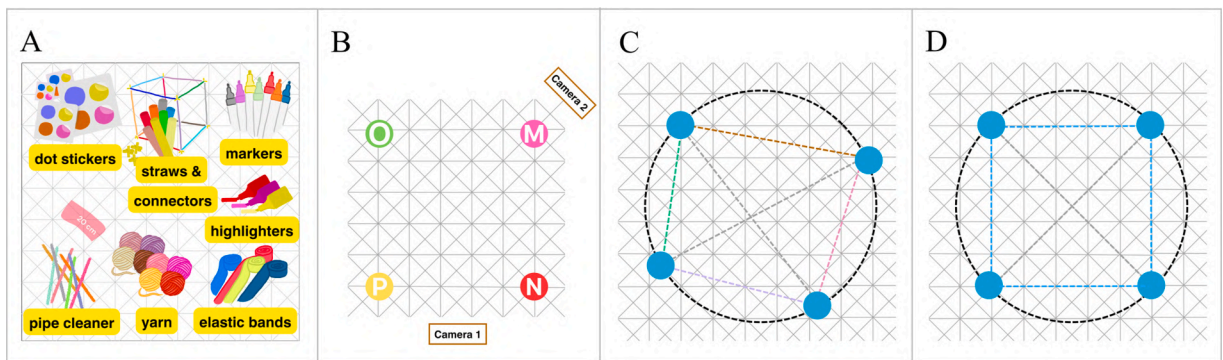


Fig. 7. The GRiD activity's material resources, data-collection setup, and hypothetical solutions to the make-a-circle task.

Note: Panel A: GRiD's components include a gridded mat and various marking accessories (Graphic artwork: Made Ariawan). Panel B: positions of the two recording cameras with respect to the group of four students, initials P, O, M, and N, standing on the mat. Panels C & D: given the collective task to form a circle, students (blue dots) could position themselves at any four points on its circumference, creating an inscribed quadrilateral (Panel C). However, the dancing context of the geometry task may implicitly encourage students to configure themselves in equidistant distribution along the circumference, spontaneously utilizing the mat's lines as auxiliary resources for the geometric construction, thus creating an inscribed square (Panel D). We hasten to note that the circle itself, which we will be using for illustrative purposes, is never actually inscribed or adumbrated but remains throughout the activity in the realm of the collective proleptic imaginary (cf. [Nemirovsky, Kelton, & Rhodehamel, 2012](#), on phantasms), multimodally referred to as a prospective indexical ([Goodwin, 1996](#)).

The study was implemented in Bali, Indonesia, with 76 fifth-grade students (aged 9–12). All students participated in a recorded 60-min teaching experiment (Steffe & Thompson, 2000). The data were analyzed qualitatively using micro-genetic ethnographic methodology focusing on moment-by-moment learning that occurred during the activity (Siegler, 2006). We attended in particular to the participants' multimodal utterances as they negotiated the collective performance of assigned tasks, following the ethnomethodological approach to conversation analysis (Mondada, 2019). For the purposes of creating unobstructed still images of the students, our data will draw alternatively from either Camera 1 or 2 (Fig. 7B), which explains discontinuities in the students' respective locations from one figure to the next.

For the purpose of investigating heteroperception, we will focus on an empirical data vignette where a group of four students were asked to configure themselves into the form of a circle. Our fifth-grade participants were familiar with the shape of a circle but had not studied its formal elements (i.e., center, radius, and diameter) let alone its form as a set of points equidistant from a center. Formally, the task can be achieved by configuring any inscribed quadrilateral (Fig. 7C). Given the activity's dance context, however, we expected the participants to spontaneously gravitate toward a special case of an inscribed quadrilateral, namely a square (Fig. 7D).

Preliminary data analysis of students' collective movement on the GRiD mat can be reduced to a rough synoptic narrative of four individuals walking silently toward each other, ultimately converging on a square configuration (Fig. 7D). Closer analysis, however, reveals that the students were each attentively micro-adjusting their positions with respect to their peers, not unlike flocking birds (cf. Wilensky & Rand, 2015), until the circle emerged and stabilized. And yet analysis of students' post-task group reflection illuminated their individual micro-adjustments as informed by different perceptions. How can four people all performing essentially the same operations (moving themselves from one grid point to another) accomplish a collective task (configuring themselves as a circle), without any plan or explicit coordination, even as they perceive their respective operations as engendering different actions? Below, we tackle this question by minutely scrutinizing the collective work of four participants: Merry (M), Nath (N), Owen (O), and Philip (P). The fourth author of this paper served as the researcher–instructor facilitating the activity (hence “teacher”).

2.2.1. Stage 1: implicit heteroperception in joint action (GRiD)

At the onset of the circle-making activity, just prior to receiving their task instructions, the four participants had been sitting on the mat, facing each other. The teacher invited them to stand up, all face the same direction (toward Camera 1 serving as the “audience,” see Fig. 7B), and walk in place in traditional dance style. Merry assumed the female style *ngegol*, while Nath, Owen, and Philip assumed the male style *mapal*. Soon after, the teacher instructed them to create a circle. The students exclaimed, “circle circle!” and, still walking, made the following adjustments: Philip and Nath, who had been facing outward of the mat center, turned to face inward toward the center of the mat, while Merry and Owen slightly rotated their bodies toward the mat center so that, in the end, Merry and Philip were facing each other from opposite corners of the mat, as were Nath and Owen, respectively (Fig. 8A). As the students were stepping in Balinese dance style to form the circle, the teacher prompted them to monitor each other and call out if they think someone is at the wrong location relative to the circle. The students, though, refrained from speaking. Instead, they gazed at each other and kept adjusting their own position (Fig. 8B–C). Once the students all claimed they were content with the circle, namely that all believed that no further adjustments are required, the teacher asked them to stand still and place a sticker to mark their location. The students responded by configuring their feet in *Tapak Sirang Pada*, a V-shape, whereby their heels met and their toes were rotated out to the corners, a posture they had previously discussed during the preceding activity (see Apsari & Abrahamson, 2026). The V-shape's vertex indexes one's location on the stage, and the students will soon be indicating that point with a sticker.

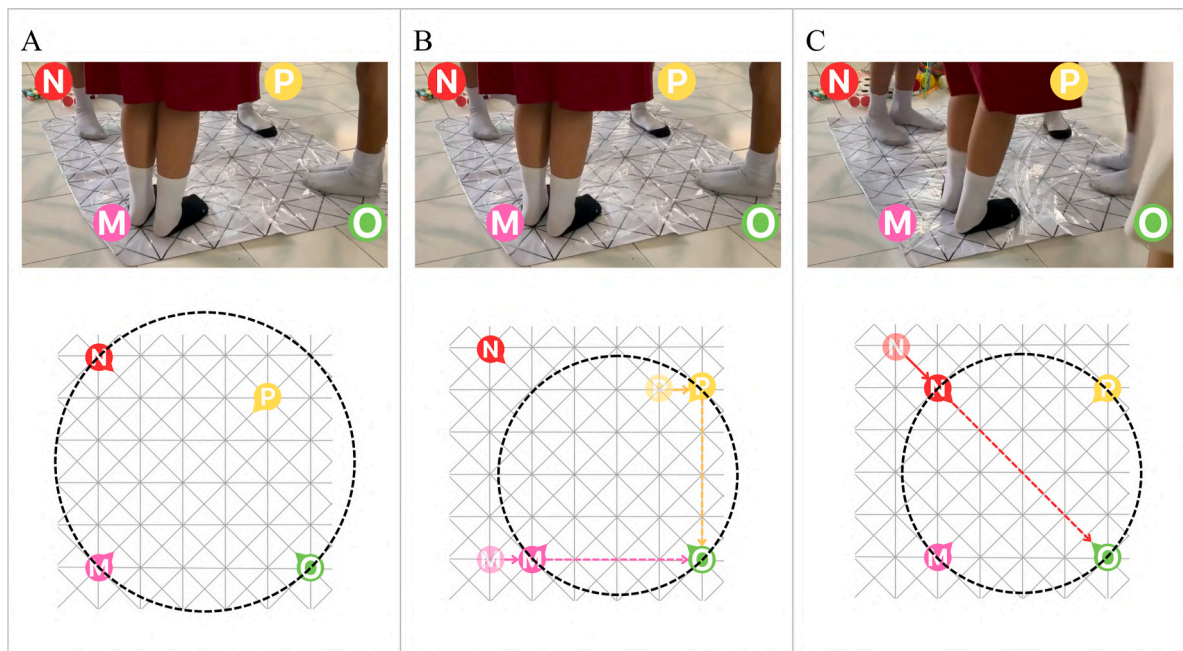


Fig. 8. Making a circle: video still shots and schematics.

Note: The students' positions on the GRiD are represented by dots with their respective initial. Little triangles protruding from each student-dot—their “noses”—indicate their respective body orientation. Faded dots indicate initial locations, solid arrows indicate their movement directions, and dashed arrows foreshadow aspectual contents of students' verbal and non-verbal explanations respecting their orientations and movements (see Sections 2.2.2 and 2.2.3, below).

Fig. 8 presents three moments in the process of Merry, Nath, and Philip adjusting their positions to form a circle. What Fig. 8 does not disclose, however, are students' visual reference points by which they planned, executed, and assessed their operations. These were revealed subsequently, during students' further interactions with each other, and later in their conversations with the teacher. These data indicate that the students were using different reference points to determine their alignment with other people in the circle. We classified their references into two types (Fig. 9): (a) *opposites*, i.e., Nath to Owen and Merry to Philip (Fig. 9A, configuring a global diagrammatic X-form); and (b) *adjacents*, i.e., Merry to each of Nath and Owen, Owen to each of Merry and Philip, Philip to each of Owen and Nath, and Nath to each of Philip and Merry (Fig. 9B; configuring global diagrammatic L-forms). Note that we will use the allocentric terms “X-form” and “L-form,” rather than capture students' egocentric orientations—“+ -form” and “V-form,” respectively—so as to illustrate through succession of figures for how local intersubjective coordination gives rise to normative geometric form.

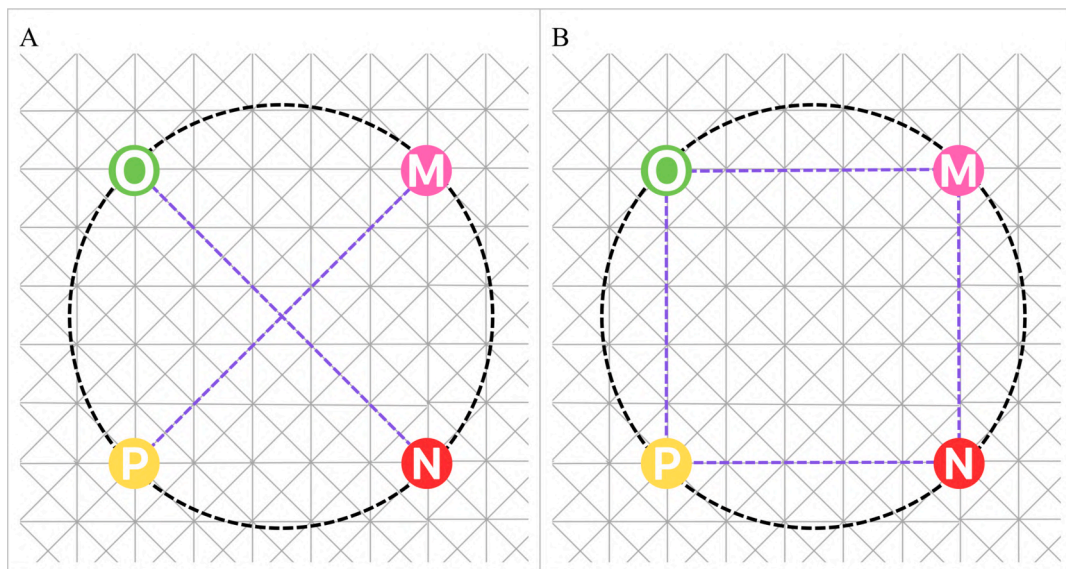


Fig. 9. Students' spontaneous strategies in forming a circle together.

Note: Dashed purple lines illustrate students' gazes as they evaluate their spatial orientation toward their peers standing either opposite (Panel A, X-form) or adjacent to (Panel B, L-forms from the perspective of each point) them.

As the students set off to mark their position with dot stickers, Owen spontaneously touched his exact location on the GRiD with his index finger calling out, "I'm here" (Fig. 10A), which then inspired the other three to do the same, either with fingers or toes (Fig. 10A-C). Yet, as they grabbed dot stickers, both Merry and Nath lost their positions on the GRiD... To re-find their respective positions, they referenced their peers' positions (Fig. 10D-F), which surfaced for our analysis a serendipitous opportunity to witness students' shared and idiosyncratic perceptions of the arena.

A  O: I'm here [points to his position with his right index finger, indicated by a green flower as an illustrative mark-up for the purposes of this transcript] P: Here I am [moves back and points his position with his left index finger, indicated by a yellow flower]	B  M: [moves aside and points to her position, indicated by a pink flower] Here N: [moves aside, points out his position with his left toe as indicated by a red flower] O: [switches to point his position with his left index finger, as indicated by a green flower, and reaches for a marker with his right hand] P: [switches to his right index finger to indicate a different point, near to his previous mark—compared to 10A—as shown by a yellow flower]	C  P: [reverting back to the initial location, as in 1a, uses his left toe to indicate the location] R: Well, now put the stickers on. M, N, O, P: Stickers, stickers. Take it, take it, take it! O: [marks his position with a marker, indicated by a green flower] P, N, M: [each grabs a sticker]
D  O: [grabs a sticker] N: Where was I? [stands, looks around] M: [obscured by Nath; uses her finger to trace a line from Owen's location towards her; highlighted here by a dashed-pink arrow] P: It has to be the same. It has to be the same.	E  O: That's why you should mark it. R: Mark it... N, M: [stand in TSP configuration; with inset image from the other camera]	F  P: [checked on Owen's position and then observed his own] I marked it before. O: I am here [places a sticker on the location he had previously marked on A–C] P: I am here [points to the location he had previously indicated with his left toe, then places a sticker there]. M & N: [place stickers on their respective positions] O: Done.

Fig. 10. Transcript of students marking their locations on the GRiD.
Transcript of Students Marking Their Locations on the GRiD [00:26:26–00:26:43]

Fig. 10D shows Merry in the act of tracing a line from Owen's position toward her, to find her own position. To pinpoint her location along this line from Owen, Merry would need to triangulate the line either with an opposite line from Philip or an adjacent line from Nath. It is the latter strategy that Merry used, we conjecture, as based on her later utterances (see Stage 3, below). At the same time, Nath, the other lost sheep, stood and peered at his peers, then adjusted his position so that he once again faced Owen, perhaps attending to a diagonal mat line emanating from Owen toward himself (Fig. 10E). Now that the students have each marked their respective locations on the circle with a dot sticker, we shall henceforth emulate the students' spontaneous practice of referring to those

four mat locations as metonymically marking the students' embodied locations.

2.2.2. Stage 2: explicit heteroperception in joint action (GRiD)

Immediately after the participants had marked their locations with colorful stickers, the teacher prompted students to explain their strategies for making a circle (Fig. 11A). Nath revealed his reference point as opposites (see Fig. 11B). Simultaneously, Philip also shared his strategy by quietly tracing a line from his point toward Owen's direction, indicating the use of *adjacents* strategy which is similar to Merry (Fig. 11B).

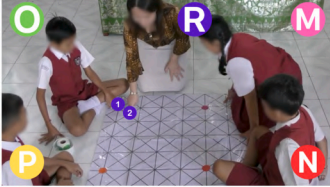
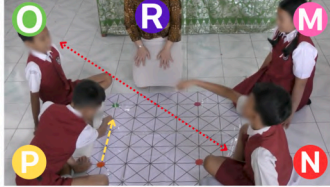
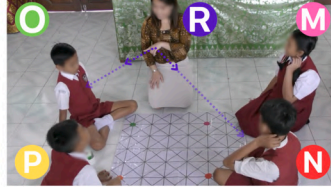
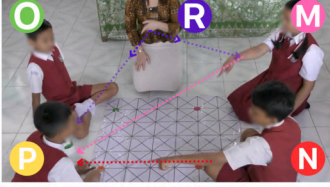
A 	B 	C 
R: Like, for example, Owen, why did you stop here? [points to 1] Why didn't you stop here instead? [points to 2]. How did you know it?	P: [gesture-traces a line on the mat from his dot-sticker position to Owen dot sticker, highlighted here by a dashed-yellow arrow] N: Face to face. Face to face with each other [gestures back and forth towards O and himself, twice, highlighted here by a dotted red double head arrow] M: [looks rapidly at Owen, then Philip, then back to Owen] O: Face to face [repeats Nath's phrase and looks at him] P: [stops tracing, looks at Nath]	R: Oh, . . . Nath looked at Owen, [simultaneously looks-and-points to Nath and then Owen, indicated with dotted purple arrows. Unlike Nath in 11B, who gestures with his arm directly above the collinear gridline, Ratih gestures with her wrist only, and these gestures are not collinear with the mat lines below her hand. Hence, her gesture is marked with a short, dotted arrow that then fades out toward its referent.]
D 	E 	F 
R, N: [simultaneously] Merry looked at Philip. [Ratih looks and points to Merry and then Philip, Nath and Merry point to Philip, Owen looks at Philip.]	P: Ehemmm [gesture-traces a line from M to P, indicated here by a dashed-yellow arrow] R: Oh, I see.	O, P: [gesture-trace a line from O to N, indicated here by a dashed-green and a dashed-yellow arrow respectively]

Fig. 11. Transcript of students explaining their orientational reference points.
Transcript of Students Explaining Their Reference Points [00:27:03–00:27:18]

Through collective semiotic activity, each person's dot stickers on the GRiD becomes metonymic of their geometrically defined embodied location, for example, Owen's sticker stands in for Owen's location (Fig. 11A, also see students' utterance on Fig. 10F, "I am here"). Responding to the teacher's prompt, Nath shared his orientational strategy vis-à-vis the person opposite to him by gesturing and uttering the phrase "face-to-face." His gesture, while pointing to Owen's body, is not a stand-alone semiotic operation. Rather, it at once invokes the grid-line connecting his sticker and Owen's sticker, as will later be elaborated by Owen and Philip (see Fig. 11E). Similarly, Merry's gesture to Philip's on Fig. 11D indicated the line between her sticker and Owen's sticker on the GRiD's mat.

Philip had been tracing a line between his and Owen's *adjacents* locations (Fig. 11B) when he first learned of the *opposites* orientational strategy ("face-to-face"). Philip stopped short still before his line had reached Owen's green dot sticker (Fig. 11B). We conjecture that Philip now becomes aware that his peer was operating under a different strategy. Similarly, Merry who was tracing the line from Owen (Fig. 11D) also demonstrated a quick realization, as is evidenced by her rapid eye movements toward Owen and Philip. Here, we evidence a moment of breakdown, when Merry and Philip first apprehended the *opposites* strategy.

Together with Merry, Philip quickly adopted Nath's *opposites*, as evidenced by Philip's pointing and tracing lines along the two diagonals (Fig. 11D–E). In addition, Owen, whose strategy had been unknown to his peers (and, later, to the analysts), claimed that he, too, had used *opposites*, presumably vis-à-vis Nath (Fig. 11B) and yet then explicitly demonstrated his endorsement of *opposites* by gesture-tracing a line between him and Nath (Fig. 11F). Overall, Fig. 11 illustrates a substitution (Benally et al., 2022) of two students' perspectives (Merry and Philip), in which one strategy (*opposites*) replaces the other (*adjacents*).

2.2.3. Stage 3: resolved heteroperception in joint action (GRiD)

Up to this point, the students had been gesture-tracing to/along the mat grid-lines to express their orientational reference points for evaluating the circle form. Looking to promote her students' pedagogically promising geometrical conversation, the teacher invited them to highlight those lines. While previously in Fig. 11D–F the students all appeared to converge on employing the *opposites* strategy, the *adjacents* are now about to make a comeback (see Fig. 12).

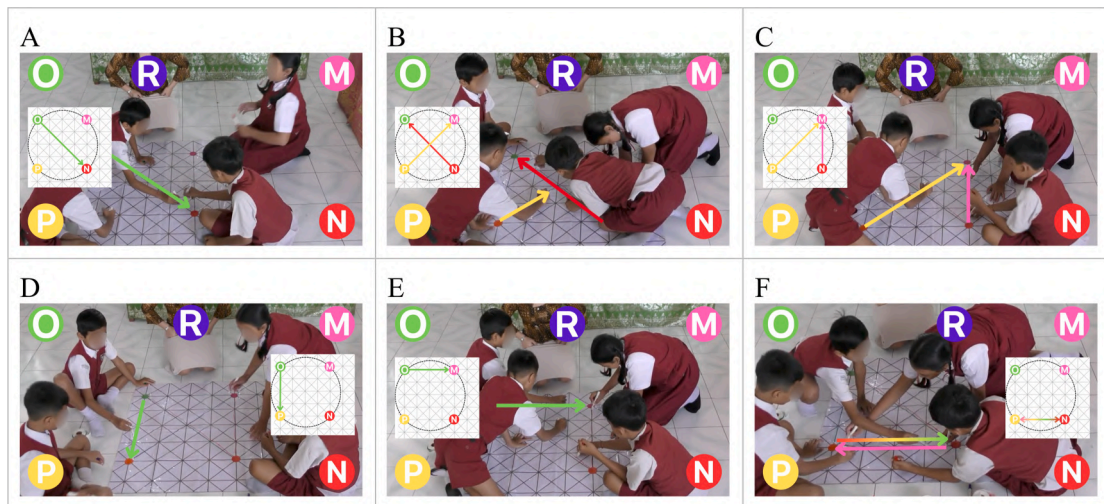


Fig. 12. Students explicitly draw their auxiliary references lines.

Note: Panel A: Owen draws a line from O to N. Panel B: Philip draws a line from P to M; Nath draws a line from N to O. Panel C: Merry draws a line from N to M; Philip continues his P–M line. Panel D: Owen draws a line from O to M. Panel E: Owen draws a line from O to P. Panel F: Merry draws a line from N to P, while everyone else draws a line from P to N. Schematics of the students' drawing on the GRiD appear as supplementary insets on each picture.

Initially, Nath, Owen, and Philip each drew a line that connects their mark to its opposite (Fig. 12A–B). However, having a chance to individually draw the lines in collaborative workspace enabled Merry to retain her initial perception. Recall that Mary had traced a line from Owen to herself (Fig. 10E). Now Merry drew a line from Nath to herself (Fig. 12C). Together, these two lines confirm that her strategy was *adjacents*. Seeing Merry draw the adjacent line, Owen drew the respective line from his location to Merry (Fig. 12D) and then to Philip (Fig. 12E). On the last panel, everyone drew the line connecting Philip's and Nath's locations (Fig. 12F). Thus, in the end, both the *opposites* and *adjacents* strategies materialized on the GRiD mat, indicating *mutuality* in perspectival negotiation (Benally et al., 2022). Having all been exposed to both perceptual orientations, as led by Owen, the students then implemented both strategies, thus achieving perspectival *synergy* (Benally et al., 2022). Fig. 13 provides a clean schematic summary of each student's line drawings on the GRiD as they picked up on each other's strategies. Specifically, Fig. 13F imbricates the cumulative auxiliary lines the group co-constructed in their collective attempt to evaluate the geometric quality of their respective locations on the collective circle formation.

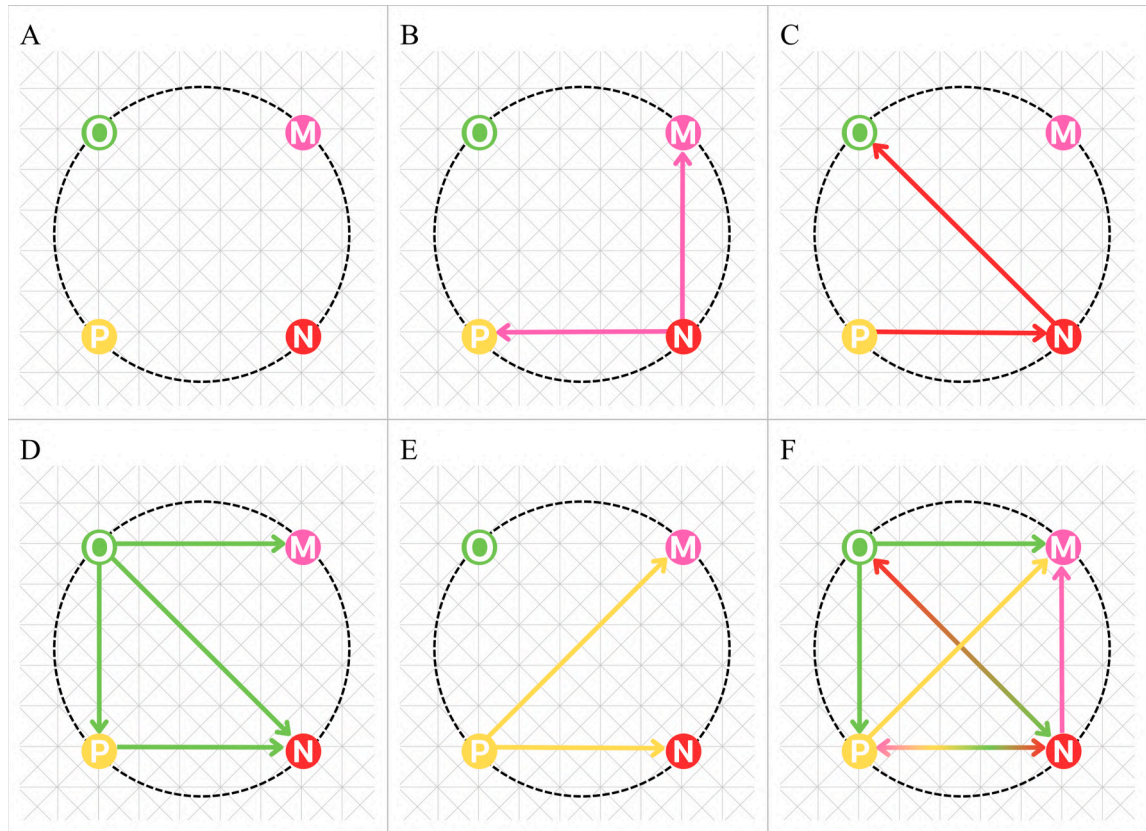


Fig. 13. Summary of students' co-constructed auxiliary lines for evaluating a circle formation.

Note: Panel A: Students' positions on the constructed circle. Panels B–F: Students' lines as indicated with their nominal colors. Panel B: Merry's lines. Panel C: Nath's lines. Panel D: Owen's lines. Panel E: Philip's lines. Panel F: Final diagram with all the lines.

By way of a summary, Fig. 14 provides schematic illustration of students' heteroperception as it evolved sequentially throughout the vignette: (1) implicit, as they formed a circle (Fig. 14A); (2) explicit, which resulted in breakdown, as the group had occasion to share their individual reference points (Fig. 14B); and (3) resolved, once shared and coordinated (Fig. 14C). As such, students' attentional anchors—their implicit perceptual structures governing the coordination of personal and interpersonal movement (Hutto & Sánchez-García, 2015)—which they each initially activated to create a group circle, became explicitly contested, once surfaced through multimodal utterance, and eventually materialized as durable signs by recruiting available semiotic resources (grid, stickers, markers), resulting in a collective “mandala” imbricating all heteroperceptions as a socially endorsed conceptual composite, that is, the normative geometric form of a square.

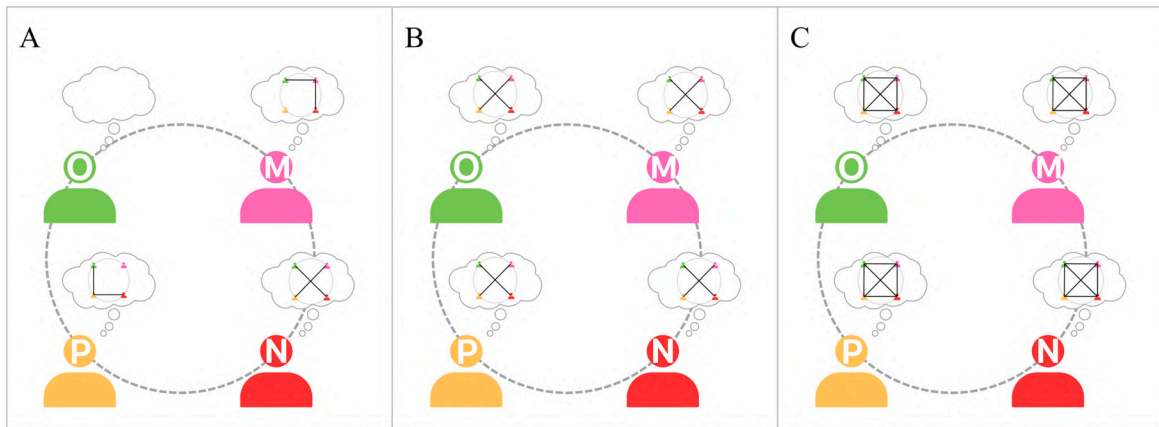


Fig. 14. From implicit heteroperception in dance practice to explicit co-constructed perception in proto-geometric discourse.

Note: Panel A: Students' initial strategies for evaluating their physical locations on the group circle, as gleaned from their gestures or verbal descriptions—they each referenced their spatial orientations relative to their peers: *opposites* (X-form) and *adjacents* (L-forms). Panel B: After the first round of discussion, all four agreed to employ *opposites*. Here, Philip and Merry substitute their initial *adjacents* with Nath's *opposites*. Panel C: Later, as students marked their reference lines, Merry reintroduced *adjacents* strategy, and her peers combined it with *opposites*, thus re-inventing the circle or, perhaps, the square.

All in all, we have evidenced that the students were initially operating based on different referential perceptions, as they created the circle (Fig. 14A). Yet, through discussion, some of the group members became aware of their implicit heteroperception and substitutes their strategy to match the others' (Fig. 14B). As the group explicitly traced their referential perception as lines in a shared diagrammatic space, the initial strategy was reintroduced, resulting in synergy between different perceptions (Fig. 14C). The two perceptions—*opposites* and *adjacents*—were never hierarchically valorized, nor was one perception better defined. Instead, the two strategies presented themselves as complementary in productively co-constraining students' collaborative embodied argumentation (cf. Palatnik, Lomos, & Abrahamson, 2026) to the effect that indeed they had made a circle.

2.3. PD by design

MJ, Cadence, and Jahlani, three elementary teachers, each with one year of teaching experience, worked for four days on iteratively designing, building, and evaluating with students a mathematics education tool. The teachers were participating in a comprehensive project evaluating a professional-development intervention oriented on teachers building content knowledge and pedagogical content knowledge through creating instructional resources (Greenstein, Fernández, & Davidson, 2019; Howiecka-Tańska, Potęga Vel Żabik, & Gop, 2023). These teachers chose to develop a versatile number-line artifact that they envisioned using as a tool for teaching several related concepts. In particular, they worked on creating a task where students would use the artifact to conceptually make sense of the fraction division statement $4\frac{1}{2} \div \frac{2}{3} = 6\frac{3}{4}$.

On Day 2 of the workshop, the teachers themselves worked through the arithmetic equation using their number-line tool as an open number-line (a blank number line where numbers and tick marks are added as needed). Initially they determined the quotient to be $6\frac{3}{4}$ by changing $4\frac{1}{2}$ into an improper fraction and then using the traditional “keep-change-change” school method, where the first fraction is kept the same, the division operation is changed to multiplication, and the second fraction is replaced with its reciprocal, that is, $4\frac{1}{2} \div \frac{2}{3}$ becomes $\frac{9}{2} \div \frac{2}{3}$ which then becomes $\frac{9}{2} \times \frac{3}{2}$. Having thus determined the solution procedurally, the teachers then proceeded to make sense of the expression conceptually by “skipping” along the number line, beginning at zero, by lengths of $\frac{2}{3}$ to determine, “How many $\frac{2}{3}$ s are in $4\frac{1}{2}$?”

The teachers found that solving the problem by hopping along the number line supported them in developing their own conceptual understanding of fraction division, and thus they decided to use the same problem for the Day 4 implementation phase of the workshop, with an adult learner. In this implementation phase, all four participants (the three teachers and the adult learner) were seated around a square table with work paper available (see Fig. 15A). The adult learner, Ali, was college educated, but self-described as “not a math person.” The teacher participants tasked Ali to “use the arrow and number line to prove that $4\frac{1}{2} \div \frac{2}{3} = 6\frac{3}{4}$ ” (see Fig. 15B). Struggling to demonstrate why $4\frac{1}{2} \div \frac{2}{3} = 6\frac{3}{4}$, Ali proposed to first make sense of the simpler division problem $8 \div 2 = 4$, with the intention of subsequently applying the structure she found there to the original problem.

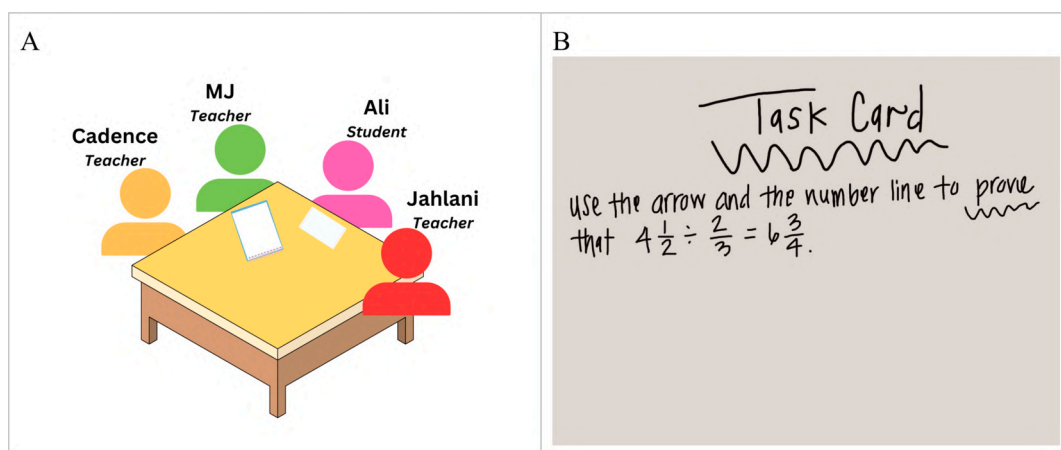


Fig. 15. Participant arrangement around table and original task card provided to the student.

At different points in the inquiry process, MJ and Ali each drew diagrams that depict a horizontal quantity—MJ's was a row of circles (see Fig. 16A), and Ali's was a line segment (see Fig. 16B)—partitioned into equally sized groups. MJ partitioned her row of circles with a vertical line, and Ali partitioned her line segment by boxing lengths of the segment (see Fig. 16A&B). Of particular interest for this paper, the diagrams were perceived differently by different participants, even though they were collaboratively discussing and referring to the diagrams. Specifically, some of the participants perceived the partition(s) as dividing the horizontal quantity into a *given number of equal groups* (for the expression $8 \div 2$, this meant that the horizontal quantity was partitioned into 2 *equal groups*, see Fig. 16C), as such the answer to the division problem was the *size of each of the equal groups* (in the case of the division problem $8 \div 2$, the answer would be that *each equal group is size 4*). In contrast, other participants perceived the partition(s) as dividing the horizontal quantity into *equal groups of a given size* (for the expression $8 \div 2$, this meant that the horizontal quantity was partitioned into *equal groups of size 2*, see Fig. 16D), so that the answer to the division problem is the *number of equal groups* created (in the case of the division problem $8 \div 2$, the answer would be that there are 2 *equal groups*). These two perceptions align to partitive and quotative models of division respectively (see Fig. 16C&D), though the participants only discovered these terms online closer to the end of their discussion.

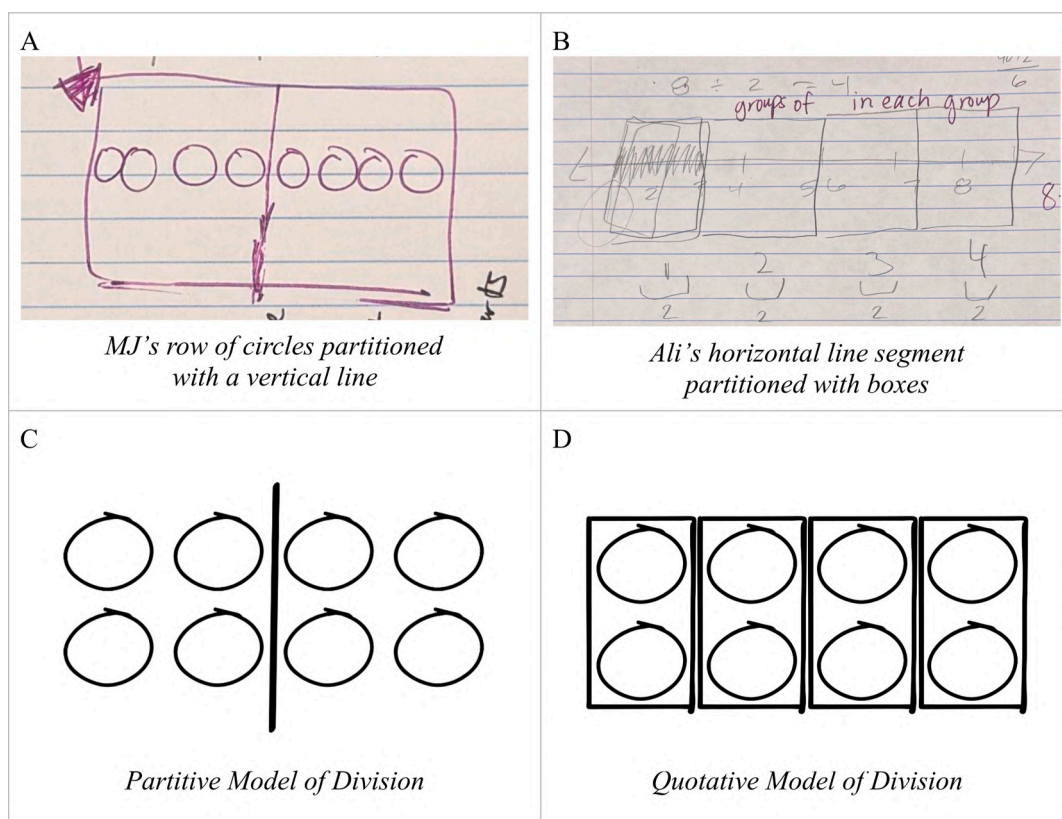


Fig. 16. Partitive and quotative division: participant work and our diagrammatic depictions.

Note: Panel A: MJ's diagram depicting $8 \div 2 = 4$. Panel B: Ali's diagram depicting $8 \div 2 = 4$. Panel C: A schematic diagram depicting the partitive model of division as applied to the statement $8 \div 2 = 4$. Panel D: A schematic diagram depicting the quotative model of division as applied to the statement $8 \div 2 = 4$. MJ and Ali's diagrams align to the partitive and quotative models of division respectively. However none of the four participants were aware of the terms "partitive division" and "quotative division" until near the end of the activity.

Once the teacher participants realized that they were operating from different perceived models of division, they tried to negotiate those perceptions with each other. They referred to both MJ and Ari's work, depicted on two separate sheets of paper, to explain their thoughts. The following analysis characterizes the proceedings along the proposed three stages of heteroperception.

2.3.1. Stage 1: implicit heteroperception in joint action (PD by design)

In the Fig. 17 transcript we see evidence that Ali is breaking 8 up into groups of 2 when she says that $8 \div 2 = 4$ means "that 2 can go into 8 four times." Additionally, looking at Ali's work from this time (see Fig. 17), Ali starts to depict $8 \div 2 = 4$ on the number line by scribbling out the 2 units from 1 to 3. This indicates that she is creating groups of length 2 (although she gets stuck and cannot immediately continue to break up eight into more sets of two).

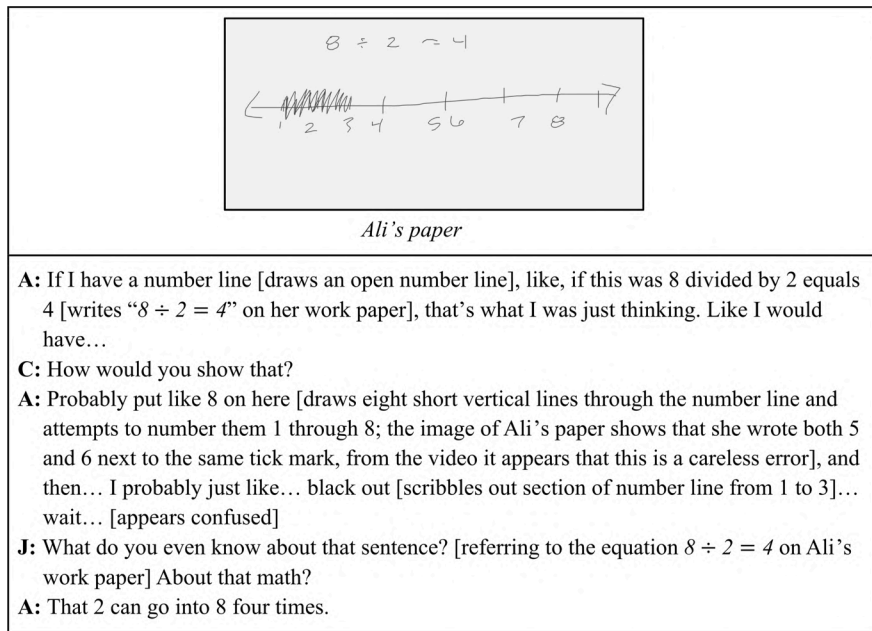


Fig. 17. Transcript of Ali demonstrating a quotative perspective.
Transcript of Ali Demonstrating a Quotative Perspective [00:25:45–00:26:53].

In the Fig. 18 transcript, Cadence and Jahlani attempt to guide Ali's thinking of the $8 \div 2 = 4$ division problem. We see that Cadence appears to also be thinking of 8 as being broken up into groups of 2. Specifically, when Cadence prompts Ali to see that 2 is the *part*, rather than the *number of parts*, she aligns to Ali's quotative model of division (additionally, when MJ and Cadence later negotiate their different perceptions, we will see further evidence that Cadence is taking a quotative perspective of division). With regard to Jahlani, she appears to reinforce Ali's quotative approach when she [Jahlani] builds on Ali's reasoning by saying "What does '2' mean? What is '2' telling you?" So there is some evidence that perhaps she also perceives the division problem according to the quotative model; however, throughout the broader conversation, it never becomes entirely clear.

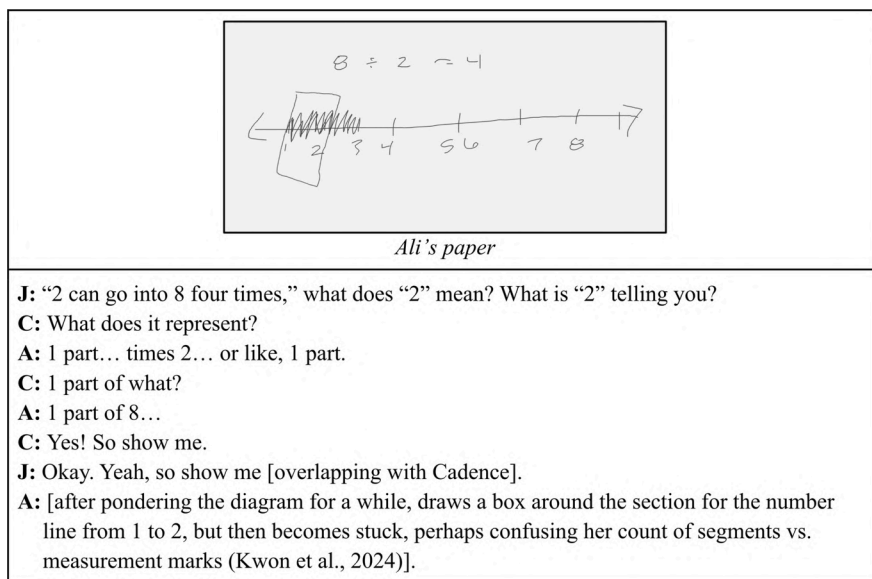


Fig. 18. Transcript of cadence aligning to Ali's quotative perspective.
Transcript of Cadence Appearing to Align to Ali's Quotative Perspective [00:26:52–00:27:54].

At the end of the Fig. 18 transcript, Ali became stuck and was uncertain how to continue partitioning her number line. In the Fig. 19 transcript, Cadence and Jahani attempt to help Ali get unstuck by prompting Ali to attend to the number of parts and the size of each part in the statement $8 \div 2 = 4$. Unfortunately, Ali becomes even more confused until MJ helps Ali reorient to the problem by asking, “[I]f you divide 8 into 2 parts, how much is *each* of those parts worth?” Importantly, MJ’s phrasing revealed that she was thinking of the division problem from the partitive perspective. This is evidenced when she first determined that there will be four parts, then went on to work out how many would be in each part.

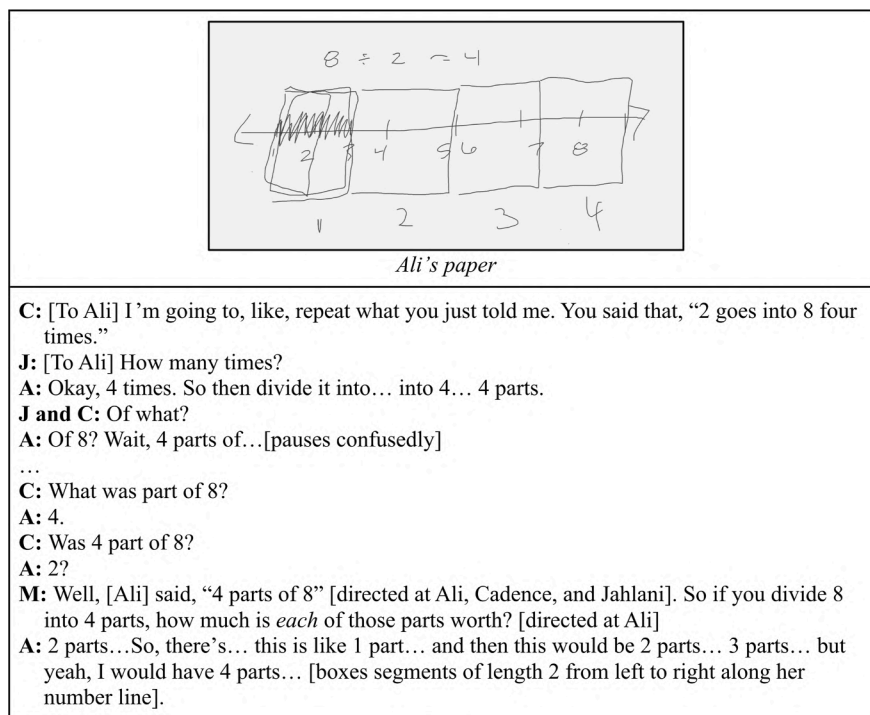


Fig. 19. Transcript of MJ demonstrating a partitive perspective.
Transcript of MJ Demonstrating a Partitive Perspective [00:27:55–00:29:04].

Even though MJ stated the partitive perspective to Ali, we see that Ali was still operating from a quotative perspective when she [Ali] boxed groups of 2 from left to right along her number line (see Fig. 19). First, Ali attended to each box being of size 2, and only at the end confirmed, "But yeah, I would have 4 parts."

These three transcripts (Figs. 17–19), exemplify implicit heteroperception within joint action. Specifically, as MJ, Cadence, Jahani, and Ali jointly engage in a teaching/learning activity where participants make sense of the $8 \div 2 = 4$ division problem, we observe that Cadence and Ali perceive the problem aligned to the quotative model of division, while MJ perceives the problem aligned to the partitive model of division (it is unclear how Jahani perceives the problem). Perhaps the participants are already experiencing some "under-the-radar" confusion, as with the use of the term "part," however, they don't yet appear to be considering whether they might be acting from different perspectives. In the next section, that confusion will build until MJ and Cadence experience breakdown and the heteroperception becomes explicit. Their current different perceptions are represented in Fig. 20.

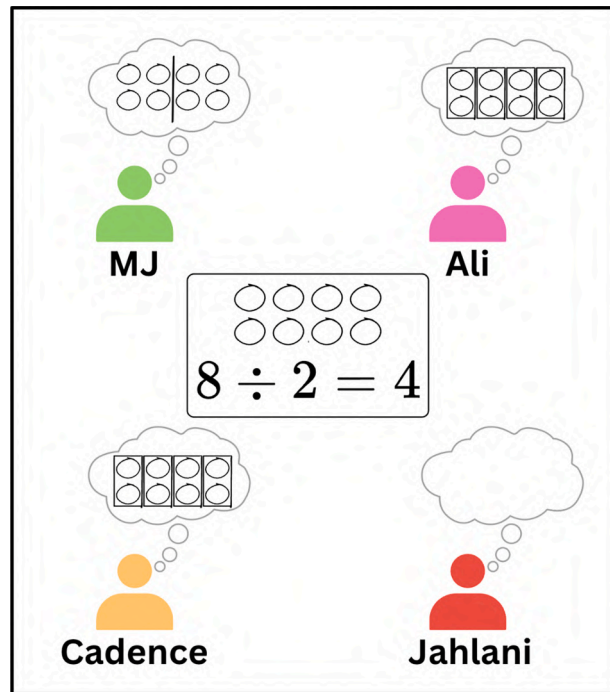


Fig. 20. Participant perceptions at the end of Stage 1.

2.3.2. Stage 2: explicit heteroperception in joint action (PD by design)

As the learning activity continued, MJ attempted to help Ali recognize her [MJ's] partitive structure of the $8 \div 2 = 4$ division problem. As she did so, we see evidence that MJ blended the partitive and quotative models of division, apparently without being aware of what she was doing. In her first utterance, MJ repeated Ali's reasoning that aligned to the quotative model (see Fig. 19). Then, in her next utterance, MJ attempted to restate the same reasoning, however she inadvertently partially applied her own partitive perspective to the statement. That is, when MJ said, "If we take 8 and divide it into groups of 2, we get... 4 in each group," her phrase at the end of the statement, "we get... 4 in each group," didn't align to the first half of the sentence, where she'd already established that the group size was 2. It appears that MJ was trying to find a way to align her and Ali's reasoning. However, she did not appear entirely conscious of what she was doing. Indeed, she didn't seem fully aware that she and Ali were operating from different perspectives, and as such, she didn't seem aware that she was attempting to reconcile those different perspectives. However, MJ did seem to be noticing that something wasn't quite right. This may have been her first inkling that heteroperception was at play.

Here, MJ started by restating Ali's perspective that aligned to the quotative model of division. Then MJ asserted that Ali's perspective was incorrect and, instead, posited that $8 \div 2 = 4$ should be thought of breaking up 8 objects into 2 equal groups of 4 (i.e., the partitive model of division). MJ drew a diagram (see Fig. 22) of 8 circles with a line splitting the circles into 2 groups of 4, to indicate her partitive perspective in contrast to Ali's quotative perspective. In that instance, MJ had become aware that she and Ali were bringing to bear different conceptual models of division. Furthermore, MJ was able to successfully communicate her [MJ's] partitive perspective to Cadence, so that Cadence also saw that two different approaches were at play. It is additionally interesting that, at this point, MJ didn't acknowledge that both models of division are mathematically acceptable. Instead, as mentioned above, MJ initially considered Ali's approach to be incorrect, and for the time being, Cadence also seems to be convinced that Ali's perspective is incorrect.

The two transcripts (see Figs. 21 & 22) show the process of MJ becoming aware that she and Ali and perceiving the $8 \div 2 = 4$ division problem differently and then pointing out the difference so that Cadence can see it as well. The participants' evolving perspectives are represented in Fig. 23.

2.3.3. Stage 3: resolved heteroperception in joint action (PD by design)

In this transcript (see Fig. 24), Cadence pushed back against MJ's partitive description of the division problem. First, Cadence described how the original $4\frac{1}{2} \div \frac{2}{3} = 6\frac{3}{4}$ problem from the task card could be thought of quotatively as, "How many sets of $\frac{2}{3}$ are in $4\frac{1}{2}$?" In response, MJ indicated openness to the idea by asking Cadence to repeat her reasoning. Cadence then goes on to connect the same quotative structure to the $8 \div 2 = 4$ division problem. The episode concluded with MJ silently thinking, apparently attempting to make sense of how both perceptions of $8 \div 2 = 4$ can be correct simultaneously (Figs. 20, 24, 27).

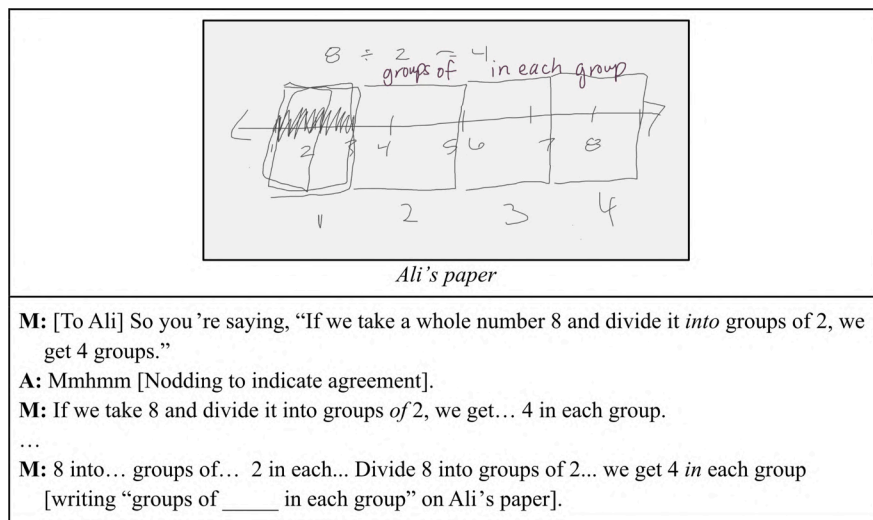


Fig. 21. Transcript of MJ combining her [MJ's] partitive model with Ali's quotative model.
Transcript of MJ Combining Her [MJ's] Partitive Model with Ali's Quotative Model [00:29:36–00:30:16].

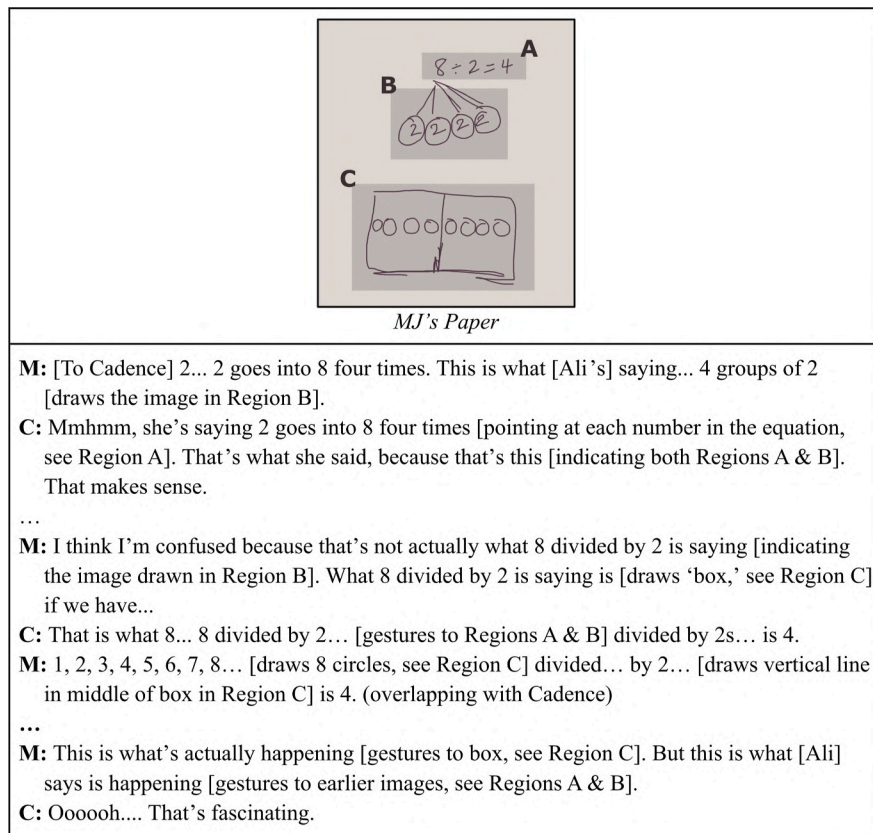


Fig. 22. Transcript of MJ becoming aware that she and Ali are perceiving the problem differently and demonstrating that difference to Cadence.
Transcript of MJ Becoming Aware that She and Ali are Perceiving the Problem Differently and Shows Cadence [00:36:24–00:37:38].

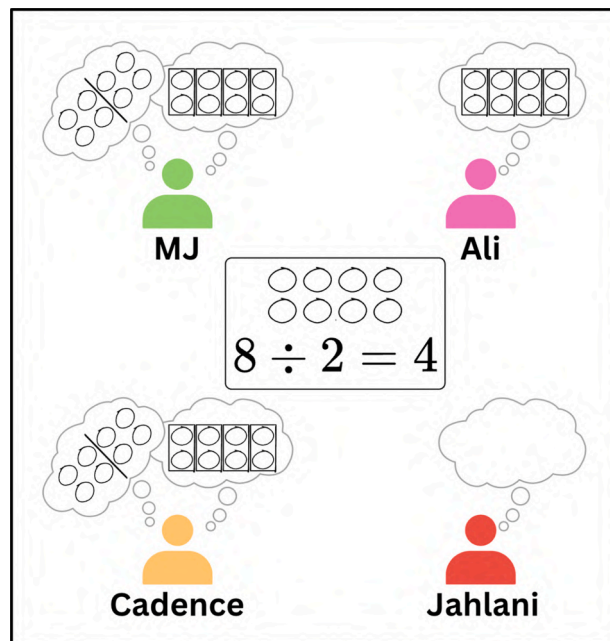


Fig. 23. Participant perceptions resulting from Stage 2.

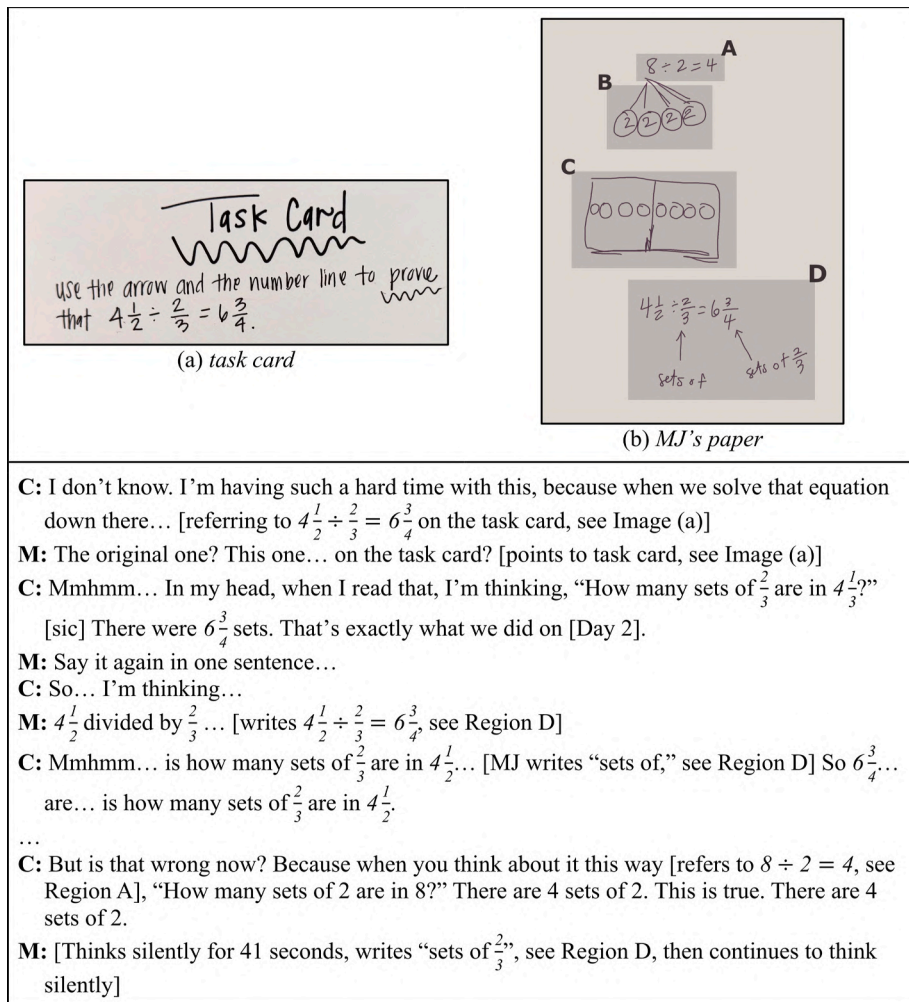
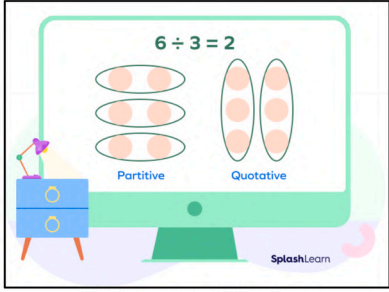


Fig. 24. Transcript of Cadence arguing that the quotative model also makes sense.

Transcript of Cadence arguing that the quotative model also makes sense [00:43:50–00:46:28].



The image on MJ's phone

M: They literally can both be right. [studying the image on her phone]. I've never heard of quotative division versus partitive division in my life [hands her phone to Cadence so she can see the image].

...

M: The only thing that sep... separates it is, "What is... what does the 2 represent?" The num... And it can be number of groups or number in each group.

...

C: The question now is...

M: Is like, groups, and number in each group.

C: [Nods in confirmation]

...

C: Okay, it makes sense now in my head why it's going back and forth.

Fig. 25. Transcript of MJ discovering that both models of division are correct.

Transcript of MJ Discovering that Both Models of Division are Correct [01:05:53–01:08:55].

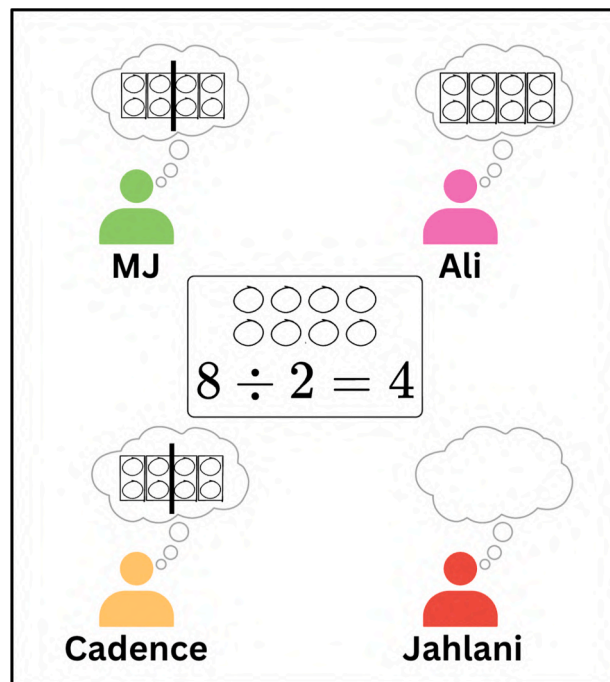


Fig. 26. Participants' perceptions after resolution.

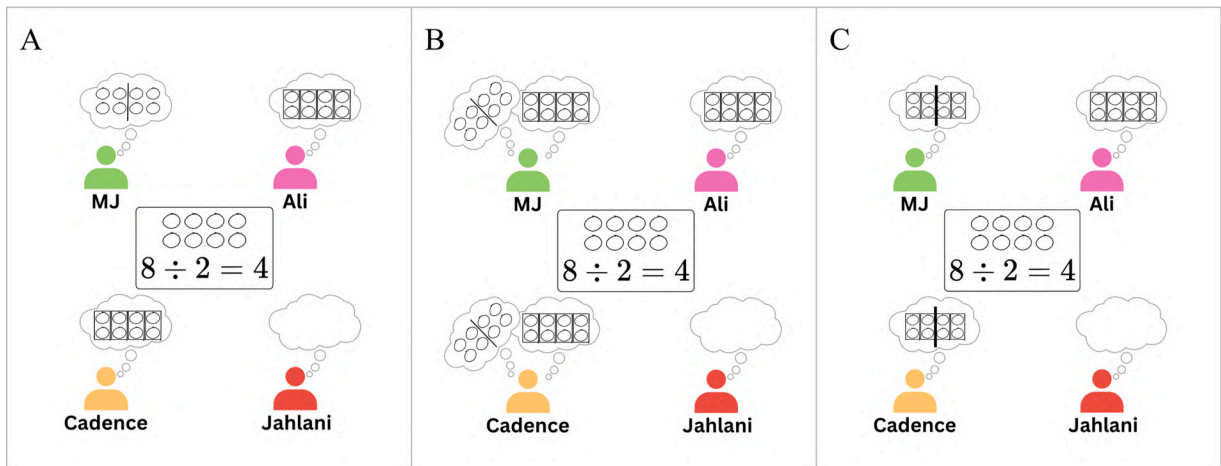


Fig. 27. Evolution of the participants' perceptions across the three stages.

MJ and Cadence went back to working with Jahlani to support Ali with the fraction division task. However, MJ still felt confused about the seemingly incompatible approaches to the $8 \div 2 = 4$ division problem. So after about 15 min, she began researching on her phone how to reconcile her and Ali's perspectives. After about 5 min of research, MJ has an aha moment while studying the image in Fig. 25. She discovered that both perspectives are correct and that they are called the partitive and quotative models, respectively. MJ passed her phone to Cadence, so that she, too, could see the image (see Fig. 25). After Cadence studied the image, she and MJ built on each other's thinking to affirm that the difference between the partitive and quotative model is that you are either being given the *number of groups* or the *number in each group*.

These two transcripts show the resolution of heteroperception as MJ and Cadence discover that not only do both perceptions of the $8 \div 2 = 4$ division problem make sense, but they are both conventional methods of division. Their final understanding of the two perceptions as compatible is represented in Fig. 26.

In summary, over the course of the three stages, we observed that MJ and Cadence started with different perceptions (partitive and quotative, respectively), but at first were unaware of their heteroperception (see Fig. 27A). As they continued to work together to support Ali, their implicit heteroperception fostered confusion around the meaning of semiotic contributions (speech referents, gestures, diagrammatic elements, etc.), which grew until breakdown resulted, eventually leading to MJ and Cadence's recognition that they were perceiving the problem differently (see Fig. 27B). Unexpectedly, MJ and Cadence's initial reaction was to consider that the two approaches could not both be correct, that is, as though the two approaches were mutually exclusive rather than, in fact, conceptually complementary. This led to an apparent tension, since both models effectively described the $8 \div 2 = 4$ division problem. Ultimately, MJ sought outside resources by conducting an internet search, where she found a diagram that supported both her and Cadence's models, presenting them as equally valorized. Specifically, the diagram prompted them to see that while the partitive model sets a required number of equivalent groups and then determines an appropriate group size, the quotative model works in an opposite fashion setting required group size, then determining an appropriate number of groups. As such, they were able to reconcile the two perceptions as synergistic mathematical models of division (see Fig. 27C).

3. Discussion

Fig. 28 summarizes our analyses of heteroperception as it manifested and evolved through the three cases. Contextual circumstances, such as the participants' age or mathematical background as well as the facilitator's interventional reach during the activity, made for certain variability across the three cases with respect to individuals' opportunities for personal growth. Notwithstanding, we hope to have demonstrated the plausibility and analytic utility of heteroperception as a theoretical construct as well as its proposed three-stage evolution.

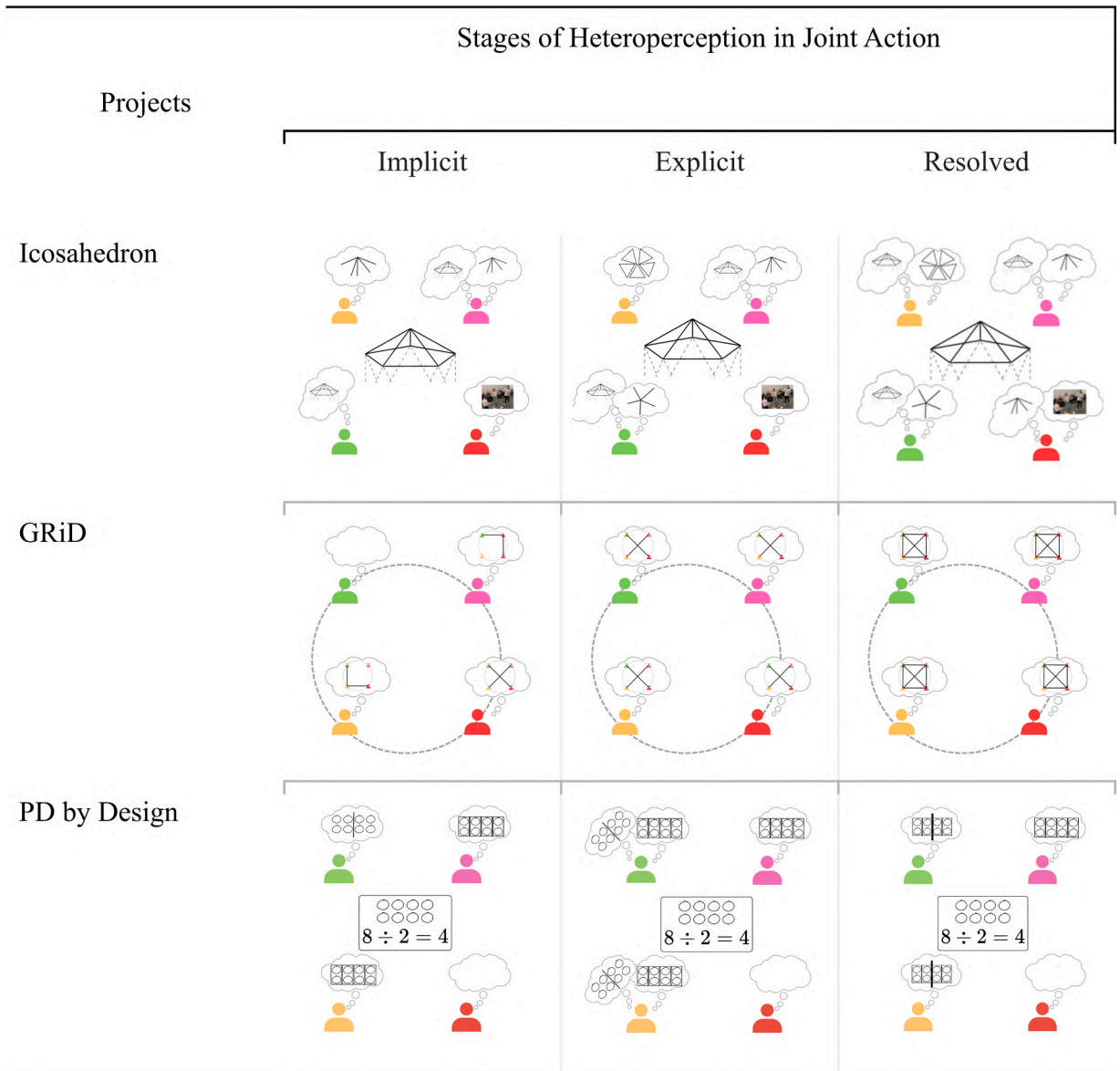


Fig. 28. Summary of heteroperception stages across the three case studies.

Articulating these proposed aggregate dynamics was gleaned in our analyses through conjecturing the reasoning process of several individuals vis-à-vis each other and then iteratively comparing across the three projects in our analytic quest to refine some semblance of a stable model that captures what we believe are universal features of human aggregates sharing a mission. Namely, our analyses have attempted to draw out the parallel structural dynamics inherent to collaborative problem-solving. We clarify that the construct of heteroperception, by its very plural constitution, takes as its unit of analysis not a single person but the asymmetries, interactions, and differential insights of a collective of individuals. Heteroperception evolves and dissolves as an iterative chain of non-linear phase shifts distributed across people, modalities, and semiotic registers. Our analytic landmarks of “implicit,” “explicit,” and “resolved” should, as such, be understood not as rigid demarcations of uniform agentive states but as gravitational points within a fluid meta-stable system. Moreover, these landmarks may obtain only to local bilateral interactions in interaction pockets whose emergent consensus may propagate to other member of the activity system, perhaps now, perhaps later, perhaps never. Yet the system as a whole bootstraps into a new functional attractor with adaptive capacity that may coalesce into new practice, for example a new mathematical discourse.

Reflecting across the three vignettes, and reconsidering some of the framing literatures, this section opens with an argument for the *necessary* role of heteroperception in collective learning, particularly in the collaborative solving of open-ended problems (Section 3.1). We then characterize heteroperception's susceptibility both to media properties and activity novelty (Section 3.2).

3.1. Naturalizing heteroperception as a vital systemic catalyst of cultural evolution

How is it possible that two or more people are engaging in what appears to be joint action even while, as it will later turn out, they are perceiving the situation differently? Across three varied data corpi, we have demonstrated the heteroperception of participants in a shared task of collaboratively creating a mathematical object by mobilizing a set of resources. These resources ranged from: (1) material construction materials; through to (2) their own bodies; and on to (3) standard stationery. At times, as they assembled a structure together, the participants literally held in their hands the very same element at the same time. How can two people perform together the task of adjoining two elements, even as they perceive the situation differently? Let us consider the icosahedron activity.

Our data made evident that B and T agreed that a particular dowel (serving as an edge) should be inserted into a particular pipe (serving as a vertex). And yet our data also made evident that B perceived this joining of the elements as completing a *roof* (pentagonal pyramid), even as T perceived this very same adjoining as completing a *five* (a total of five edges at the vertex). As such, B and T's dissonant mental orientations toward the shared display appear to belie their consonant manual behaviors on this display. At least, we submit, these flagrant dissensus/consensus data may invite us to ponder whether we should think of B and T as engaged in "joint action," for in what sense could this action be joint when each perceives it as a different action?

Perhaps a way out of this Gordian knot is to stratify the notion of an action prismatically such that it might accommodate different perceptions. While we have deliberately avoided centering Leont'ev's framework in our theoretical background, our findings resonate with his model of activity, particularly its hierarchical structure of activity–action–operation (Leontiev, 1981). Participants in our vignettes appear to share an overarching *motive*—engaging in an organized *activity* pertaining to a mathematical topic (e.g., the icosahedron, squareness, or division)—yet they harbor divergent goal-directed *actions* ("roof" vs. "five") while exhibiting mostly similar *operations* (joining the dowel–edge to the pipe–vertex).

We propose to conceptualize an *action* as a behavior that realizes cognitive intentionality to instantiate an end-goal, such as to generate a spatially constituted form, whether material or imaginary (Abrahamson, 2026). In turn, an action is achieved by actuating lower-order *operations*. As such, B and T could be in explicit consonance over their *co-operation* of fitting a specific dowel in a specific pipe even as they are in implicit dissonance over the greater *action* they are thus actuating (roof vs. five). Unlike cultural–historical activities with a rich discourse supporting the practice of performing actions, this novel pedagogical activity publicizes its motive and suggests its operations but deliberately does not a priori disclose its actions. The relative rarity of discovery-based collaborative activities in our cultural horizons (e.g., escape rooms) may mean that participants do not have ready dispositions to anticipate heteroperception or fathom its operative implications for coordinating joint action.

An implicit divergence of actions, and therefore of perceptions, is surfaced when participants' respective operations are in conflict. Such breakdowns surface the "local history" of the activity—an emergent amalgam of semiotic traces, partial insights, and improvised configurations that dynamically evolve and iteratively shape both the context for and constraints on subsequent actions. This micro-sociogenetic view of collective activity aligns closely with the construct of a *substrate* in co-operative action (Goodwin, 2013): a discursively cumulative interweaving of residual traces from individual contributions to a group's evolving adaptive orientation toward some shared domain. Relatedly, Palatnik & Abrahamson (2022, 2026) elaborate a networked synthesis of two accounts for joint action—embodied perceptual learning and cognitive-anthropology—arguing that heterogeneous sensorimotor percepts become publicly shared knowledge through multimodal coordination, as individual know-how sediments into a semiotic substrate that scaffolds an emerging collective "professional vision." The substrate is built not on perceptual uniformity but explicitly on successive, multimodal, *heteroglossic* semiotic elaborations (cf. Bakhtin, 1981). In this sense, heteroperception is not a procedural flaw in collective action to be precluded or suppressed, neither does it mark some of the perceptions as ultimate misconceptions, rather, heteroperception is an essential condition of collaboration, and moreover in joint explorative problem-solving under conditions of uncertainty due to limited information in unfamiliar spaces. Under such conditions, initial heteroperception mobilizes favorable systemic dynamics. When surfaced, heteroperception may cause interactional misalignments, yet once repaired, it may promote the collective formation of a new cultural form—a mutually construed adaptive perception–action grip on the situation at hand; a better way of getting things done. Heteroperception is a feature, not a bug, and this feature can be simulated and stimulated in classroom socio-genesis of cooperative problem solving.

3.2. Media, signs, and social interaction: implications for educational design

The three vignettes differed with respect to each activity's available media, that is, the activity's construction tools or modeling materials. Our analysis suggests that these respective media afforded the participants to engage in different semiotic modes:

- (a) In the icosahedron activity, the media are concrete materials—wooden dowels and vinyl connectors, and each participant manifests their intentionality by adjoining or disjoining the dowels to form geometric edges subtending vertices;
- (b) in the GRiD activity, the media include the participants' own bodies as marking locations and configuring collective shapes, where standing and moving instantiate one's role in the geometric structure; and
- (c) in the arithmetic activity, shared information is displayed in the form of diagrammatic inscriptions on paper, where what you create bespeaks your reasoning.

Of immediate note are differences in spatiality. Whereas the icosahedron and GRiD activities involve three-dimensional voluminous forms (a Platonic solid, human bodies moving across the floor), the arithmetic activity transpires for the most on two-dimensional surfaces (pen and paper). Plausibly, 3D objects, more so than 2D objects, lend themselves to multiple perspectives—as you revolve around them, different images of 3D objects fall on the retina. Still, the long literature on ambiguous 2D images attests to their capacity to trigger different perceptions Abrahamson, Lee, Negrete and Gutiérrez, 2014. On the other hand, a plurality of perspectives per se

need not guarantee heteroperception—a rose is a rose by any other perspective.

Across the three activities, participants engaged in collaborative argumentation that drew on multiple expressive modalities (e.g., speech, gesture). However, comparing across the activities we discern in participants' constructed artifacts different gradients of expository and rhetorical content, which we ascribe to the existential constitution of the different phenomena in question:

(a) The icosahedron construction, a palpable material artifact, did not lend itself to generating durable diagrammatic signs on the structure itself—the shared domain of scrutiny consisted only of vertices and edges with no surplus signage, where no supplementary semiotic resources were offered (e.g., yarn, pipe cleaners);

(b) The GRiD geometry-dance activity included a variety of semiotic resources for registering geometric information onto the mat surface, including the participants' bodies as embodying data points, stickers to tag selected locations, and colour markers to trace auxiliary lines marking significant spatial relations between locations; yet.

(c) In the arithmetic activity, at least in the vignette in question, there was no single *a priori* shared visual display—no body-scale polyhedral structure, no gridded mat, and no eight apples or what-have-you that needed to be divided by 2—and so participants relied even more so on pen-and-paper, with each person having to create anew their own diagram on the paper, unless they were embellishing each other's diagrams.

Which brings us to the following speculation about diagrams.

The diagrams generated by participants in the arithmetic activity are interesting inscriptions, at least when treated conceptually through the lens of semiotic analysis. Let us consider again the partitive and quotative models of 8 divided by 2. Note, in Fig. 16, how both images comprise *eight circles* and yet the images differ with respect to the *linear elements signifying the circles' groupings*. We wish to submit that, as a comprehensive diagrammatic construction, the circles constitute the shared domain of scrutiny—the “raw” *phenomenon* we are all gathered to confer over—yet the lines manifest *how we are orienting perceptually toward this shared phenomenon*. The diagram is an inscribed composite of ‘what’ I am looking at and ‘how’ I am looking at it—it is both the ambiguous figure (shared domain of scrutiny) and the duck or the rabbit (highlighted embedded Gestalt). Thus, the diagrammatic images are not just the shared domain of scrutiny—they also implicitly carry a particular perspective. Put simply, when people draw circles and then either put them in boxes or draw a line between them, they are both saying, “Here are circles” (the phenomenon) and, “This is how I am seeing the circles” (overlaid auxiliary markings intended to express one's idiosyncratic mathematical model of the phenomenon). These additional markings are intended to carry for our interlocutors similar discursive and semiotic portent of spoken or gestured signs that “change the environment” (Shvarts & Abrahamson, 2019). The supplementary markings inscribe our implicit attentional anchors as explicit auxiliary lines (Apsari & Abrahamson, 2026). Our idiosyncratic perceptual orientation is instantiated in the form of diagrammatic elements that parse the phenomenon into conceptually meaningful elements. Circles are the things, lines are how I am seeing the things. Inscriptions, as epistemic forms (Collins & Ferguson, 1993), mobilize the epistemic game of surfacing tacit phenomenology (Bamberger, 2020; Donald, 2010; Kirsh, 2010; Olson, 1994).

By carrying a *this-is-how-I-see-it* portent on top of merely *this-is-it*, individually generated diagrams (as in the arithmetic activity) thus instantiate an interlocutor's perspective perhaps more cogently than in the case of material forms alone (icosahedron) or embodied forms on a shared spatial arena (GRiD). As such, diagrams may more urgently precipitate collaborative argumentation that, in turn, is more likely to precipitate productive breakdown, because diagrams are inherently showing not only, “Here's the thing, okay, are we all happy?,” but also “Here's the thing *as I see it*.” Diagrams disambiguate regions of the shared domain of scrutiny by carrying the individual's pragmatic or rhetorical message, explicitizing heteroperception. And so, diagrams are composed of a shared domain of scrutiny overlaid with highlighting and often coding as well. Diagrams bear the potential of rendering covert heteroperception in the form of overt idiosyncratic representations, thus mobilizing conceptual discourse (Bamberger, 2011; diSessa, Hammer, Sherin, & Kolpakowski, 1991). Diagrams thus afford collaborative argumentation, because they inhere idiosyncratic perspective. Once interlocutors are aware of *how* they are each making sense of a display, they are both stimulated and equipped to enter in the negotiation of meanings (Asterhan & Schwarz, 2009), as may often be the case in teachers' endeavors to shift students' attention (Mason, 1989) toward ideal forms (Abrahamson et al., 2025; Flood et al., 2020; Shvarts & Abrahamson, 2019) or the classroom's hegemonic discourse (Sfard, 2007). Diagrams thus invite productive epistemic conflict.

These reflections on media, signs, and social interaction may prove useful for pedagogical theory and practice by way of demonstrating the semiotic affordances of different resources and activity configurations for collaborative argumentation. For example, if an educator wishes to precipitate productive breakdown in construction-oriented collaborative activity involving concrete materials and/or the participants' bodies, analogous to our icosahedron or GRiD activities, they would do well to include in the learning environment supplementary semiotic resources by which individuals could overlay the ‘how’ of their perception on its ‘what.’ Furthermore, educators could look to interleave diagrammatic argumentation into the activities, where each participant is motivated to recreate from scratch the objects in question, as in our arithmetic activity. Interleaving diagrammatic argumentation into collaborative activities could perhaps be achieved by introducing interaction constraints that stimulate the participants to render explicit their perspectives, reasoning, inferences, planning, or prediction (e.g., see Potega vel Žabik, Abrahamson, & Ilowiecka-Tańska, 2024, for a case of leveling up digital constraints in a dyadic motor-control coordination task).

Our tentative speculations on heteroperception's instructional utility bespeaks a need for further research into best pedagogical practices for leveraging this collective phenomenon as a learning resource. When might heteroperception be inherently unproductive (“...let's call the whole thing off”)? How might the students' age affect their acceptance of heteroperception (cf. Gopnik & Rosati, 2001)? How might the teacher's intervention facilitate or inhibit productive heteroperception (Bamberger, 2010)?

4. Conclusions and future directions of research

When a group of people solve a mathematical problem together, they may not see eye to eye, even when their collaborative joint action is effective in performing task objectives. Rather, their idiosyncratic perceptions of the situated models, solution enactments, and semiotic displays may be—perhaps necessarily are—heterogeneous. Yet, even though mathematics teachers often employ collaborative problem solving, educational research has barely documented students' heteroperception or investigated its dynamics and pedagogical tradeoffs (but see Palatnik et al., 2026). We have demonstrated across three different contexts how four participants in a collaborative activity exhibited three coordination phases pertaining to their perception of a shared situation: (1) *implicit heteroperception*—they operated together without realizing they were perceiving the situation differently; (2) *explicit heteroperception*—as collective circumstances shifted, their behaviors manifested and thus communicated their different perspectives, causing breakdown; and ultimately (3) *resolved heteroperception*—they negotiated and reconciled their idiosyncratic views, adjusting their perspectives, goals, or understandings so as to realign a functional synergy moving forward.

Collaborating agents' behavioral dynamics from heteroperception to functional synergy can be productively modeled by adopting Leont'ev's distinction between lower-level *operations*, their composite *actions*, and overall *activities*. By this hierarchical model, collaboration under heteroperception can be accounted for by noting that participants in a joint activity may tacitly agree over a specific operation even as they each perceive it as achieving a different action.

Our findings suggest that heteroperception—the condition in which learners engaged in joint activity perceive shared objects or goals differently—is not an anomaly to be avoided, but a natural, even generative, feature of collaborative mathematics learning. As movement-based and spatially grounded pedagogies gain traction, heteroperception is likely to become more visible in classrooms. Consider a future in which non-sedentary, dynamic learning environments become mainstream. Teachers should therefore be prepared not only to recognize the emergence of heteroperception but to leverage it intentionally as a pedagogical resource.

The endorsement of heteroperception as a familiar construct in educational parlance, among practitioners as well as researchers, could usher in classroom adoption of activities deliberately designed to elicit heteroperception along with facilitations guidelines (Abrahamson, 2006; Abrahamson & Wilensky, 2007; Benally et al., 2022). That said, the productive endorsement of heteroperception into classroom discourse may introduce facilitation challenges. Some students may feel vulnerable when their view of a shared situation appears less sophisticated than their peers' views (i.e., on a lower social rung of professional vision; Goodwin, 1994). As such, productively endorsing heteroperception into classroom discourse requires teachers a priori to foster environments where divergent perspectives are normalized and students are encouraged to compare, reflect on, challenge, and reconcile differing viewpoints through collaborative argumentation (Asterhan & Schwarz, 2009; Glachan & Light, 1982). Specifically, teachers could endorse an awareness, valorization, even celebration of heteroperception into what Feucht (2010) calls the classroom's *epistemic climate*, an implicit codex that moderates students' attitudes toward knowledge—its sources, forms, and discursive norms (see also Yackel & Cobb, 1996). Recognizing that their individual voices are welcome, students may experience greater agency as mathematical thinkers (cf. Heyd-Metzuyanim & Sfard, 2012). In turn, classroom adoption of these activities could create empirical contexts for evaluating and theorizing facilitation methodologies that normalize heteroperception as an anticipated feature of complex group activities. More broadly, research should evaluate for any tradeoffs of implementing heteroperception, such as the difficulty some teachers may experience in productively orchestrating heteroglossia. In parallel, longitudinal research could evaluate the effect of experiencing heteroperception on students' collaborative epistemic practice. Do they become more tolerant for ambiguity? Do they function better as team players, such as by accepting and trusting a peer who sees things differently? And what forms will heteroperception take among students of diverse sensorimotor capacity, such as blind or visually impaired students?

As teachers come to acknowledge the prevalence of heteroperception, they will require methodologies for surfacing, interfacing, and reconciling perspectival differences. Supporting this shift will require investments in teacher education. Professional learning opportunities should help teachers recognize and respond to heteroperception in real time, particularly in embodied, emergent activities. One promising approach is the use of multimodal learning analytics (MMLA)—such as mobile eye-tracking—that enable teachers to engage in collaborative tasks from learners' multiple points of view and reflect on divergent gaze patterns and attentional foci (see Palatnik et al., 2026). Such tools can help educators experience the three-stage process of heteroperception—implicit, explicit, and resolved—from the inside out, fostering a deeper awareness of its dynamics, complexities, and affordances.

In parallel, as teachers, teacher educators, and researchers cultivate productive orientations toward the potential pedagogical function of heteroperception, it should be strategic for them to understand the affordances of different activity genres, in particular different media, for foregrounding students' personal construals of the problem space. Educational designers of learning activities where students physically enact their mathematical solution propositions could consider incorporating semiotic functionalities that solicit and retain traces of individual students' thinking process, thus surfacing implicit heteroperception even when it has not disrupted effective joint action.

Finally, a single qualitative investigation, even one that compares across three different studies, comes with significant limitations. Whereas our case-study vignettes were cherry-picked to lend credence to our argument, and therefore we cannot assert the prevalence of these phenomena, we speculate that our analyses will hold quite broadly, if not ubiquitously, whenever people get together to do something new involving the utilization of familiar resources toward accomplishing unfamiliar goals. As we prepared this manuscript, in fact, our most productive moments were those where we suddenly realized that we were perceiving the data differently.

CRedit authorship contribution statement

Dor Abrahamson: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project

administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Catalina Lomos:** Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sam Hilkey:** Writing – original draft, Visualization, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ratih A. Apsari:** Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alik Palatnik:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

The authors do not have permission to share data.

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